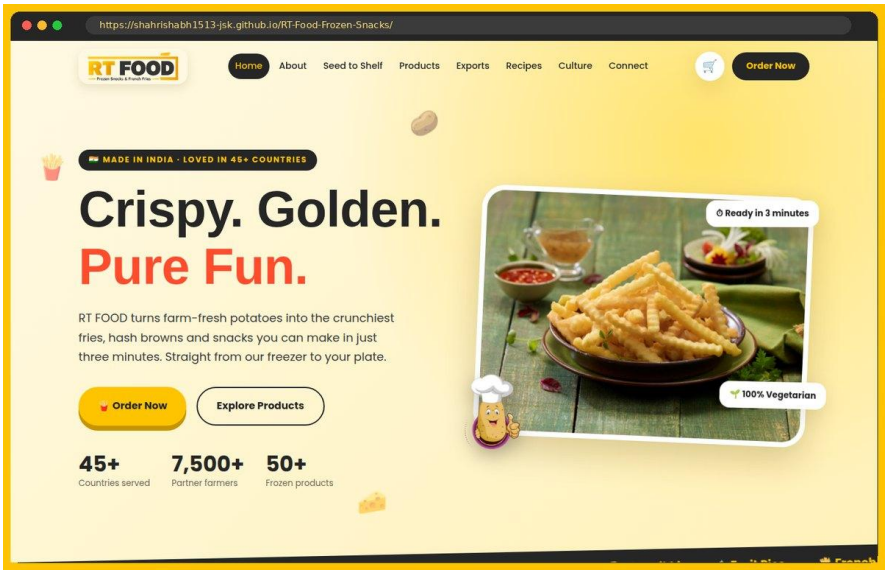
A PERSONAL PROJECT REPORT

RT FOOD
Frozen Snacks & French Fries

FROZEN SNACKS & FRENCH FRIES

WEBSITE · ONLINE ORDERING · LIVE TRACKING

A multi-page brand website with product catalogue, coupons, payment, PDF billing and live rider tracking – built with HTML, CSS and JavaScript



52	9	46	33
FROZEN PRODUCTS	WEBSITE PAGES	EXPORT COUNTRIES	VIDEO RECIPES

Project Title	RT FOOD – Frozen Snacks & French Fries Website and Ordering Portal
Technologies	HTML5 · CSS3 · JavaScript (ES6+) · localStorage · Leaflet · jsPDF · QR code · OSRM
Project Type	Personal project (front-end only, no backend) – educational demo
Developer ID	RT1504
Live Project	shahrishabh1513-jsk.github.io/RT-Food-Frozen-Snacks
Source Code	github.com/shahrishabh1513-jsk/RT-Food-Frozen-Snacks
GitHub	github.com/shahrishabh1513-jsk
LinkedIn	linkedin.com/in/rishabh-alpeshabhai-shah

DESIGNED AND DEVELOPED BY

SHAH RISHABH ALPESHBHAI

PROJECT AT A GLANCE *one page summary*

RT FOOD is a complete, fully animated brand website for a frozen-snacks and French-fries company. Visitors browse **52 products**, pick a pack size, apply coupons, pay through a **UPI QR, card or cash-on-delivery** demo checkout, download a **PDF invoice** and watch a rider move on a **live map**. It uses only HTML, CSS and JavaScript, runs on GitHub Pages and keeps cart and orders in the browser.

CUSTOMER SIDE

- Browse 52 products in 6 categories
- Search, filter, sort, pick pack size
- Add to cart or **Buy Now**
- Coupons, GST and delivery fee
- Pay, get bill, track rider live

BUSINESS SIDE

- Export page with world map
- 46 countries, 9 regions
- Export enquiry and contact form
- Offices map, FAQ
- Careers page with CV upload

EXPERIENCE

- Animated loader and hero
- 33 video recipes + PDF cards
- Seed-to-shelf and About story
- Responsive on phone and desktop
- One shared design system

LIVE CART

Cart saved in localStorage and shared across open tabs

5 COUPONS

WELCOMERT, FLAT100, FRYDAY, CRISPY50, FREESHIP

PDF DOCUMENTS

Invoice and recipe cards made in the browser with jsPDF

LIVE TRACKING

Rider moves along a real OSRM road route on a map

WORLD EXPORT MAP

Click a pin to see the shipping route from Mehsana

16 / 16 TESTS

Automated browser checks passed on the project code

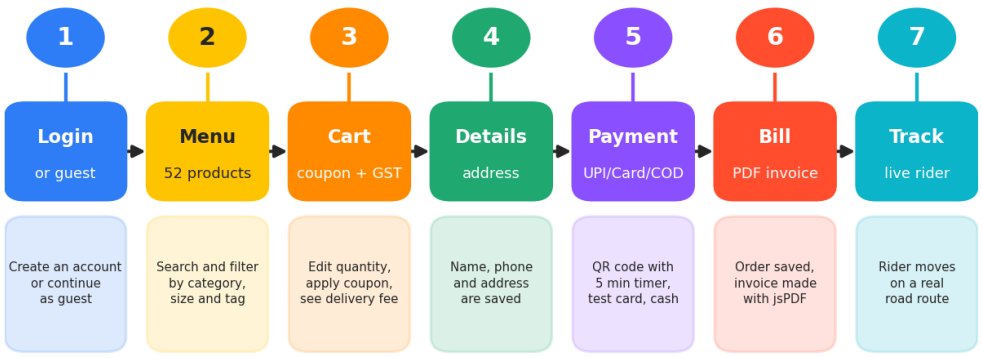
1 Browse menu → **2** Add to cart → **3** Pay online → **4** Get PDF bill → **5** Track rider

HOW IT WORKS

HTML5 | CSS3 | JavaScript ES6+ | localStorage | Leaflet | jsPDF | QR code | GitHub Pages

WORKFLOW DIAGRAM

Figure: End-to-end customer ordering workflow



TRY THE DEMO ORDER FLOW

Coupon: `WELCOMERT` (10% off, max Rs 150)

Test card: `4242 4242 4242 4242`, any future date

Delivery: free above Rs 499, else Rs 49

Payments are simulated – no real money is charged.

QUICK LINKS

Live: Live Project

Code: Project Repository

GitHub: GitHub Profile

LinkedIn: LinkedIn Profile

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CERTIFICATE

This is to certify that **SHAH RISHABH ALPESHBHAI** has successfully completed the personal project entitled **"RT FOOD – Frozen Snacks & French Fries: A Multi-Page Website with Online Ordering and Live Tracking Using HTML, CSS and JavaScript"**.

The project was designed, developed, tested and documented by the candidate using HTML5, CSS3 and Vanilla JavaScript. The work presented in this report is original and has been carried out sincerely and to the best of the candidate's ability.

This report is submitted as a record of the candidate's personal project work.

_____

Project Guide

(Name & Signature)

Date: _____

_____

Head of Department

(Name & Signature)

Place: _____

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported me during the planning, design, development and documentation of my personal project, **RT FOOD – Frozen Snacks & French Fries**.

I am thankful to my project guide and the faculty members of my department for their valuable guidance, encouragement and constructive suggestions, which helped me improve the quality of this work. Their feedback motivated me to focus on clean code, a smooth ordering journey and a modern, attractive user interface.

I also thank my family and friends for their constant support, patience and motivation throughout the project. I am grateful to the open web community and to the creators of freely available learning resources and open-source tools such as MDN Web Docs, Leaflet, jsPDF, qrcode.js, OpenStreetMap and OSRM, which made it possible to learn and apply modern web technologies effectively.

Finally, I thank everyone who directly or indirectly helped me in completing this project successfully.

SHAH RISHABH ALPESHBHAI

Developer ID: RT1504

ABSTRACT

RT FOOD is a modern, fully animated multi-page website for a frozen-snacks and French-fries brand. It brings the brand story, the complete product range, an online ordering system, export enquiries, recipes and company contact details into one responsive website. The project is developed using only **HTML5, CSS3 and Vanilla JavaScript (ES6+)**, without any framework, build step, backend server or database, and it is published as a live website on GitHub Pages.

The website has **nine connected pages**: Home, About, Seed to Shelf, Products, Exports, Recipes, Culture, Connect and Order Now. The product catalogue contains **52 products in six categories** with pack sizes, tags, search, filters and sorting. A customer can add items to the cart or use Buy Now, apply one of five coupons, see GST and delivery charges, enter delivery details, pay through a UPI QR code, a card or cash-on-delivery demonstration gateway, download a PDF invoice and follow the order on a live map where a rider moves along a real road route.

The cart, login, delivery details and orders are stored in the browser with localStorage and shared between open tabs. Invoices and recipe cards are generated as PDF files with jsPDF, payment QR codes are generated in the browser, and maps use Leaflet with OpenStreetMap data, geocoding and OSRM routing. The Exports page lists **46 countries in 9 regions** with an enquiry form, and the Recipes page offers **33 video recipes** with text steps and downloadable cards.

A set of 16 automated browser checks – page loading, catalogue rendering, filters, pricing, cart storage and mobile layout – all passed. The project demonstrates DOM manipulation, state management, event-driven design, document generation, map integration and responsive UI design.

Keywords

RT FOOD, Frozen Snacks, French Fries, HTML5, CSS3, JavaScript, localStorage, Leaflet, jsPDF, E-commerce Workflow, Live Order Tracking, Responsive Design

CHAPTER 1

INTRODUCTION

1.1 Project Overview

RT FOOD is a browser-based brand and ordering website for a frozen-snacks company. The company turns farm-fresh potatoes into French fries, hash browns, wedges, patties, pizzas, puffets, pies and momos that can be cooked in about three minutes straight from the freezer. This project presents that brand online in a bright, golden and playful design and lets a visitor complete a full order without leaving the website.

The project is written only with HTML5, CSS3 and Vanilla JavaScript. There is no server and no database: the cart, login, delivery details and orders are stored in the browser's localStorage, and open tabs stay in sync. The site is published on GitHub Pages, so it opens on any device with a modern browser.

Main highlights of the project:

- **Nine-page website** – Home, About, Seed to Shelf, Products, Exports, Recipes, Culture, Connect and Order Now – all sharing one design system.
- **52 products in 6 categories** with pack sizes (200 g, 500 g, 1 kg, 2.5 kg and 1 / 2 / 4 packs), tags, search, filters and sorting.
- **Complete order journey** – Login, Menu, Cart, Details, Payment, Bill and live Track.
- **Five coupons**, 12% GST, and a delivery fee that becomes free above Rs 499.
- **UPI QR, card and cash-on-delivery** demonstration payments with a refreshing QR timer.
- **PDF invoices and recipe cards** generated in the browser, and a **live rider map** with real road routing.
- **33 video recipes**, an export world map with 46 countries, a careers form and an offices map.

> **AT A GLANCE**
> 9 pages · 52 products · 6 categories · 33 recipes · 46 export countries · 5 coupons · 7-step order journey · 16 automated checks passed.

1.2 Problem Statement

Many small and mid-size food brands have a website that only shows photos and a phone number. Customers cannot compare pack sizes, check prices, apply offers or follow an order, so every sale depends on phone calls and messages. Business buyers and exporters also struggle to find clear information about the company's range, certifications and export reach.

Large e-commerce platforms solve this but are expensive, heavy and not designed around a single brand story. There is a need for a simple, free and fast website that presents the brand attractively, lets customers order and track in a few clicks, and also serves business visitors – while remaining a hands-on learning project for modern front-end development.

1.3 Objectives of the Project

- To design a bright, attractive and fully animated brand website using only front-end web technologies.
- To present the full product range with pack sizes, prices, tags and powerful search and filters.
- To build a complete order journey – cart, coupons, GST, payment, bill and live tracking.
- To generate PDF invoices and recipe cards directly in the browser.
- To tell the brand story through About, Seed to Shelf, Culture and Exports pages.

- To keep one consistent design system and one shared header, footer and cart for all pages.
- To make the site responsive for phone, tablet and desktop, and to test it with automated browser checks.

1.4 Scope of the Project

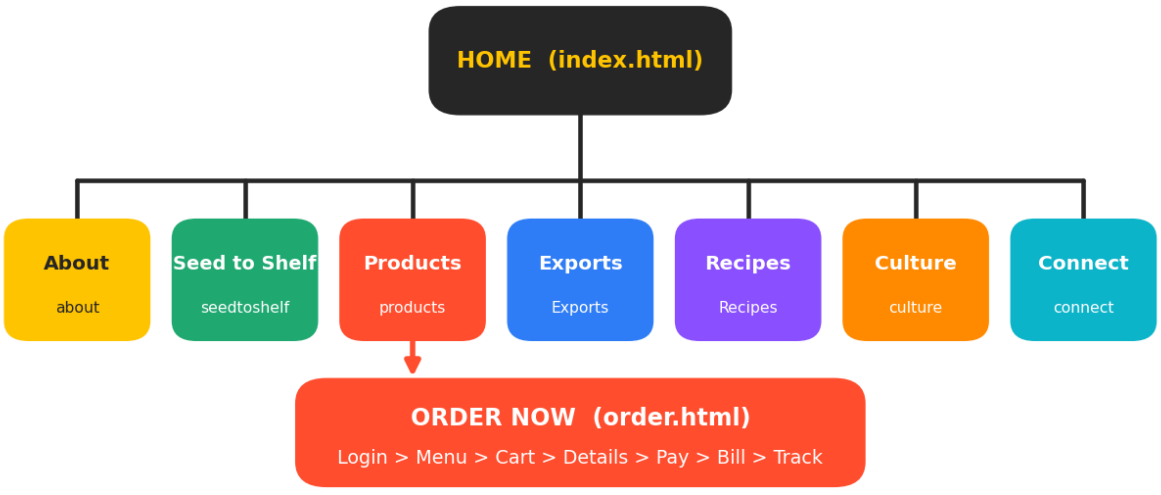
Included in the scope: brand pages, product catalogue with filters, cart and coupons, demo login and guest checkout, delivery details, UPI / card / cash payment demonstration, PDF invoice, live order tracking with map, 33 recipes with PDF cards, export map and enquiry form, careers form, contact page with FAQ, and deployment on GitHub Pages.

Not included: a real server or database, real payment-gateway integration, secure password storage, e-mail / SMS delivery and real rider dispatch. Payments, accounts and rider movement are simulated for learning. These are listed as future enhancements in Chapter 8.

1.5 Users and Roles

Role	Main purpose	Typical actions
Customer	Buys frozen snacks online and follows the order.	Browse products, filter, add to cart, apply coupon, pay, download bill, track rider, watch recipes.
Guest	Orders quickly without creating an account.	Choose “continue as guest”, fill delivery details and place an order.
Business / Exporter	Looks for bulk, private-label or export supply.	Read the Exports page, view the world map, send an export enquiry, contact sales.
Job applicant	Wants to join the RT FOOD team.	Read the Culture page, choose a department and upload a CV.
Owner / Admin	Maintains the content and prices.	Edit products, prices, coupons, GST and UPI ID in one file (js/data.js).

Figure: Site map - 9 connected pages



All pages share one design system (css/style.css) and one header, footer and cart (main.js)

Figure 1.1: Site map of the RT FOOD website

CHAPTER 2

TOOLS AND TECHNOLOGIES

2.1 HTML5

HTML5 gives the structure of every page. The project has nine HTML pages – index, about, seedtoshelf, products, Exports, Recipes, culture, connect and order. Each page contains the page-specific content, while the header, footer, cart drawer and loader are injected by one shared script so that all pages stay consistent.

- Semantic structure for header, navigation, sections, forms and footer.
- Form controls such as text, e-mail, phone, select, file upload and checkboxes for orders, enquiries and CV upload.
- Embedded YouTube videos for recipes and Leaflet map containers for tracking, exports and offices.

2.2 CSS3

One shared stylesheet, `css/style.css` (about 2,600 lines), controls the colours, layout, animation and responsive behaviour of all nine pages. The look is warm and playful: golden yellow, charcoal black, tomato red and cream, with rounded cards, soft shadows and a chunky 3D-style button.

Design token	Colour	Use
--y	#FFC400 (golden yellow)	Primary brand colour, buttons, highlights
--ink	#262626 (charcoal)	Text, header buttons, dark sections
--cream / --cream2	#FFF9E3 / #FFF1B8	Page and card backgrounds
--red	#FF4D2E (tomato)	Accent words, offers, warnings
--green / --blue	#1FA971 / #2F7DF6	Success states, vegetarian badge, links

- CSS variables so the whole theme can change in one place.
- Flexbox and Grid for the header, product grid, order stepper and footer.
- Keyframe animations for the loader, floating emojis, marquee strip, counters and confetti.
- Media queries that turn the navigation into a slide-in mobile menu.

2.3 JavaScript (ES6+)

JavaScript is the brain of the website. It builds the product cards, filters and sorts the catalogue, runs the cart, calculates prices, handles login and payment, generates PDFs and QR codes, draws maps and moves the rider. All code lives in small files under `js/`, and everything is attached to one global namespace object `RT`.

File	Responsibility
data.js	Single source of truth: products, categories, prices, pack-size multipliers, coupons, GST, export countries and 33 recipes.
main.js	Loader, header, footer, cart drawer, toast messages, scroll animations and counters.
shop.js	Product card template, pack-size chips, add-to-cart / buy-now and the `filterProducts` function.
home.js / products.js	Home-page bestsellers and the Products page filters.
order.js	The full order flow: login, menu, cart, coupons, details, payment, QR, bill and tracking.

File	Responsibility
pdf.js	Recipe-card PDF and invoice PDF using jsPDF.
recipes.js / exports.js / forms.js	Recipe browser with videos, export world map and enquiry, CV and contact forms.
logo.js	Logo stored as base64 so it can be drawn inside generated PDFs.

2.4 Browser Storage and Tab Sync

There is no database. The cart (`rtfood_cart`), login session, delivery details and placed orders are stored in the browser's localStorage as JSON, so they remain after the page is refreshed. Because the browser fires a storage event in the other open tabs, the cart count and order status update everywhere without reloading.

2.5 External Libraries and Services

Library / service	Version	Why it is used
Leaflet	1.9.4	Interactive maps for order tracking, export world map and office locations.
jsPDF	2.5.1	Creates the PDF invoice and the recipe cards in the browser.
qrcode.js	1.0.0	Draws the UPI payment QR code from the UPI ID and amount.
OpenStreetMap / CARTO tiles	–	Free map tiles and address search (geocoding).
OSRM	–	Real road routing so the rider follows an actual route to the customer.
YouTube embed	–	Plays the recipe videos.
Google Fonts	–	Web fonts for headings and body text.

2.6 Development and Deployment Tools

- **Git and GitHub** – version control and the public repository.
- **GitHub Pages** – free static hosting for the live site.
- **Python simple web server** (`python -m http.server`) – to run the site locally.
- **Modern web browser developer tools** – debugging, responsive testing and storage inspection.

CHAPTER 3

PROJECT DESIGN AND MODULES

3.1 System Architecture

RT FOOD follows a simple layered design. The presentation layer is plain HTML and CSS; the application layer is a set of small JavaScript files sharing the `RT` namespace; the services layer uses browser storage and a few open libraries; and the whole site is hosted as static files on GitHub Pages.

Figure: Layered system architecture of RT FOOD

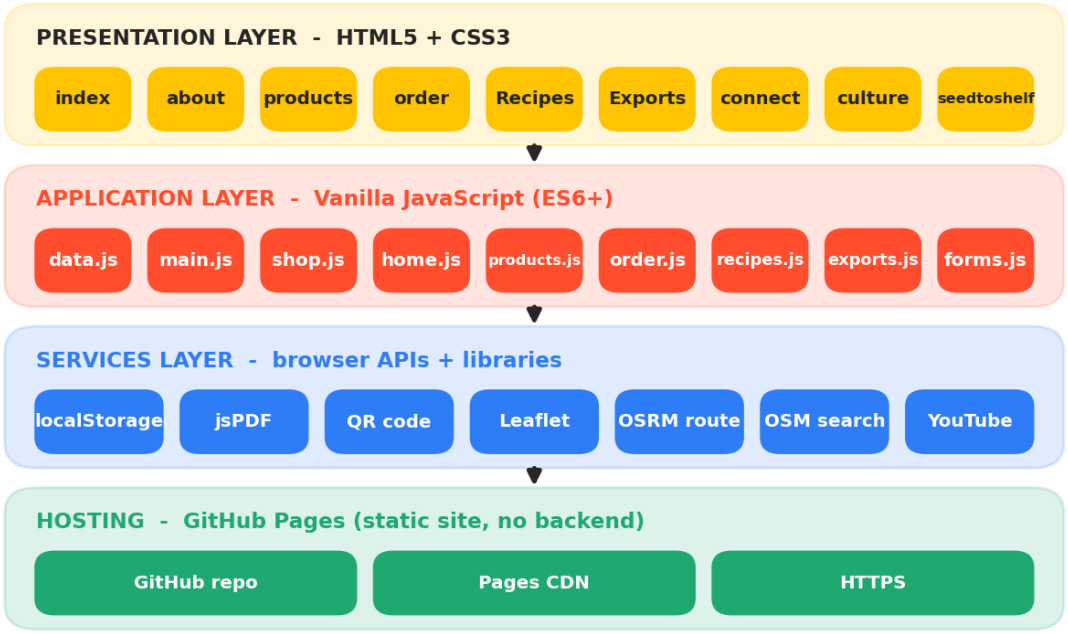


Figure 3.1: Layered system architecture

3.2 Pages and Navigation

Every page has the same header with the logo, seven menu links, a cart button and an Order Now button, and the same footer with contact details. The table below explains the purpose of each page.

Page	Purpose	Key features
Home	First impression and bestsellers	Animated hero, 45+ countries counter, scrolling category strip, bestseller cards, recipe teasers
About	Company story	Timeline from 1962 to today, mission, vision, farmer and sustainability values
Seed to Shelf	Supply chain	Eight-step journey, partner farmers, cold storage and quality standards
Products	Full catalogue	Search, category, pack-size, tag and sort filters; add to cart and Buy Now
Exports	Business buyers	World map with 46 countries, region filter, export enquiry form
Recipes	Cooking ideas	33 video recipes, text steps, PDF recipe card download
Culture	Careers	Company values and CV upload form with department choice

Page	Purpose	Key features
Connect	Contact	Phone, e-mail, three office addresses, message form and FAQ
Order Now	Online ordering	Seven-step order flow with payment, bill and live tracking

3.3 Product Catalogue Module

All products are listed in `data.js` as short arrays – category, name, icon, description and tag. At start-up they are converted into product objects with an id, base price, image and the pack sizes that suit the category. Fries and potato items are sold by weight (200 g, 500 g, 1 kg, 2.5 kg); baked snacks, pizzas, puffets and pies are sold as 1, 2 or 4 packs.

Figure: Product catalogue by category

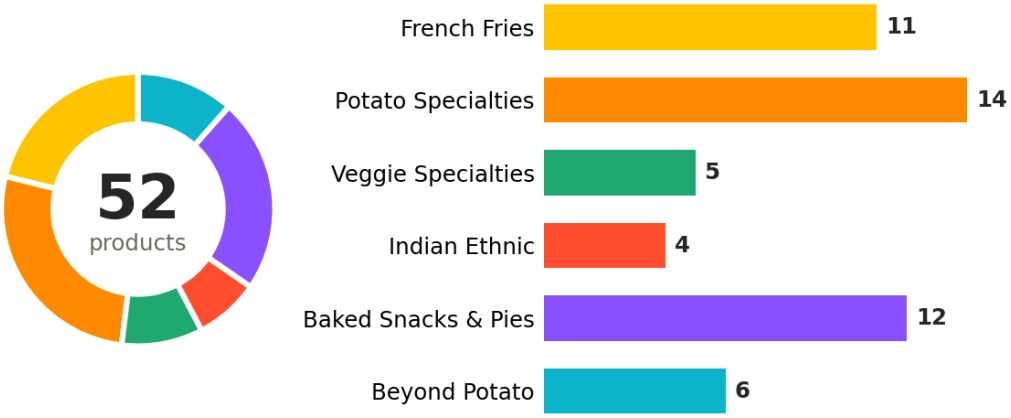


Figure 3.2: Product catalogue by category

Category	Products	500 g base price	Examples
French Fries	11	Rs 149	Crinkle Fries, XLF Fries, Peri-Peri Fries, Cajun Fries
Potato Specialties	14	Rs 159	Hash Brown Triangle, Wedges, Tater Shotz, Potatobets
Veggie Specialties	5	Rs 179	Jalapeño Cheesy Pops, Veggie Stix, Kings Patty
Indian Ethnic	4	Rs 169	Aloo Tikki, Sabudana Patty, Hara Bhara Kebab
Baked Snacks & Pies	12	Rs 199	Margherita Pizza, Schezwan Puffets, Apple Pie
Beyond Potato	6	Rs 189	Mozzarella Cheese Stix, Onion Rings, Momos

Pack sizes change the price by a multiplier on the 500 g base price: 200 g × 0.42, 500 g × 1, 1 kg × 1.9, 2.5 kg × 4.5, and 1 pack × 1, 2 packs × 1.9, 4 packs × 3.6. Product tags – Bestseller, New, Spicy, Kids Fav and Vrat Friendly – can be used as filters.

3.4 Cart and Coupon Module

The cart stores each product with its chosen pack size and quantity. On the cart step the subtotal is calculated, a coupon can be applied if its minimum order is met, delivery is added unless the order is above Rs 499, and 12% GST is added to get the grand total.

Figure: Cart pricing and coupon logic

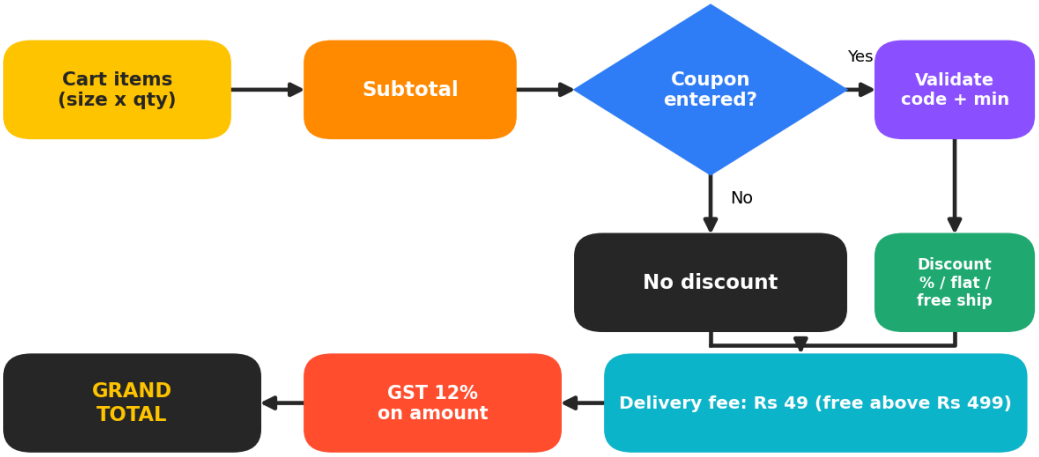


Figure 3.3: Cart pricing and coupon logic

Coupon	Benefit	Condition
WELCOMERT	10% off, maximum Rs 150	Welcome offer, no minimum
FLAT100	Flat Rs 100 off	Order above Rs 699
FRYDAY	15% off, maximum Rs 200	Order above Rs 999
CRISPY50	Flat Rs 50 off	Order above Rs 349
FREESHIP	Free delivery	Any order

3.5 Ordering Workflow

The Order Now page walks the customer through seven steps shown as a progress tracker at the top: Login, Menu, Cart, Details, Payment, Bill and Track. The customer can move back and forth, and the cart is never lost because it is saved in the browser.

Figure: End-to-end customer ordering workflow

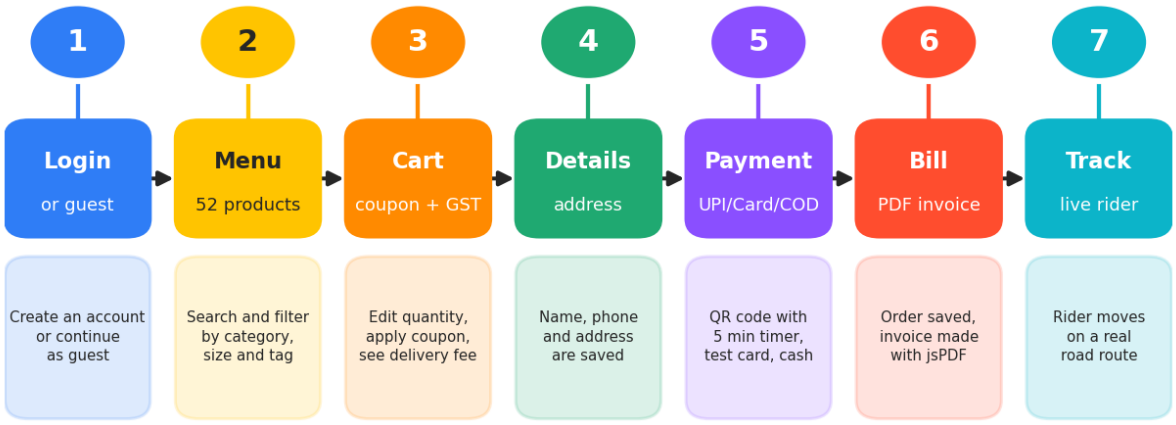


Figure 3.4: End-to-end customer ordering workflow

Login, sign-up and guest checkout

A returning customer logs in, a new customer signs up with name, phone, e-mail and password, and anyone in a hurry can continue as a guest. Demo accounts are stored only in the browser.

Figure: Login, sign-up and guest checkout flow

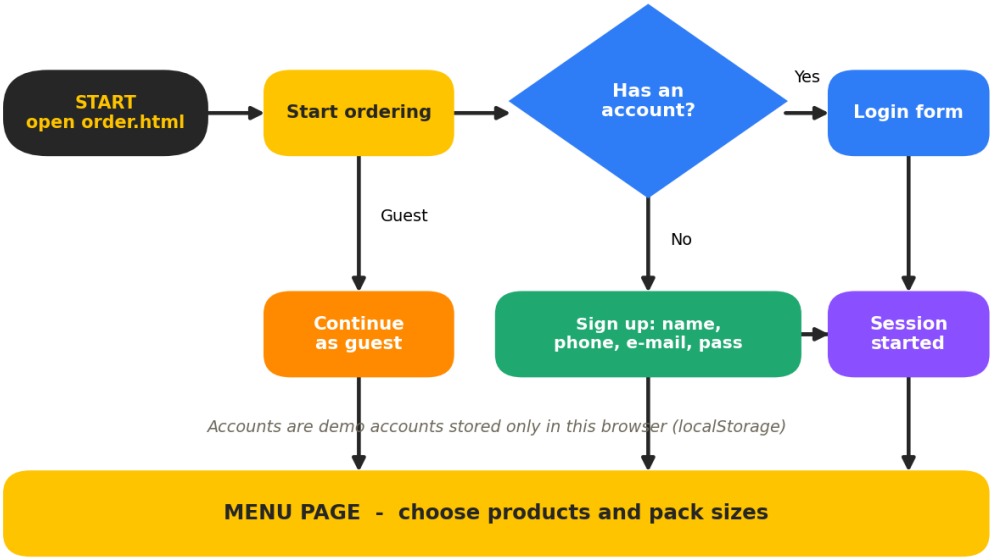
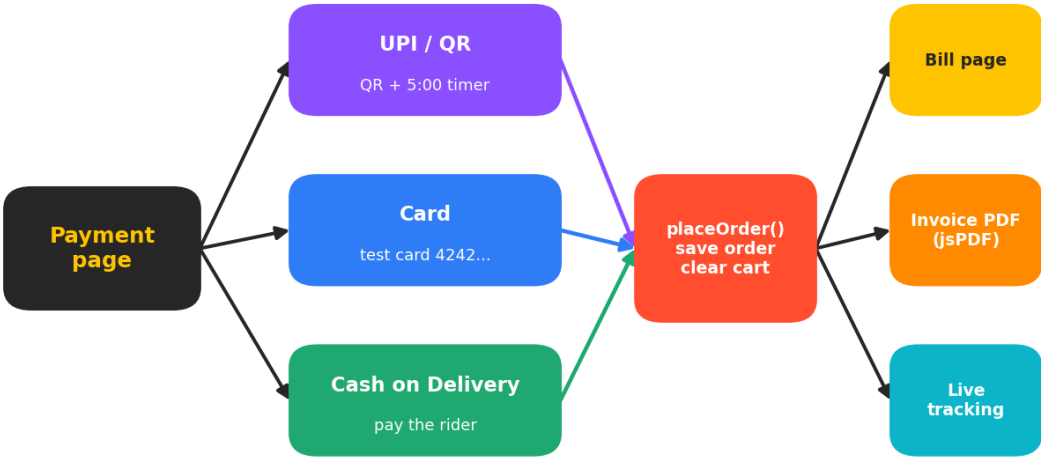


Figure 3.5: Login, sign-up and guest checkout flow

3.6 Payment Module

The payment step offers three methods. **UPI / QR** draws a QR code for the exact amount that refreshes every five minutes, with an option to open the UPI app. **Card** accepts the demo number 4242 4242 4242 4242 with any future date. **Cash on Delivery** asks the customer to keep the exact amount ready. When the order is placed it is saved, the cart is cleared and the bill page opens.

Figure: Payment methods and order placement flow



Payments are a simulated demo - no real money is charged

Figure 3.6: Payment methods and order placement flow

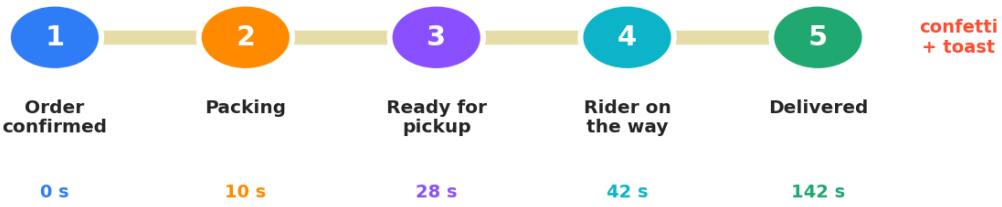
Demo only

No real payment is processed. For real sales the `placeOrder()` function would be connected to a backend and a payment gateway such as Razorpay.

3.7 Live Order Tracking

After the order is placed the customer sees the status move through five stages. The delivery address is converted to map coordinates with OpenStreetMap geocoding, a road route from the Thaltej (Ahmedabad) dispatch hub is fetched from OSRM, and a rider icon moves along that real route. When the order is delivered, confetti and a toast message appear. The timeline is compressed to about two minutes so that it can be demonstrated quickly.

Figure: Live order-tracking timeline



Demo timeline is compressed (TRACK in js/order.js). Route: OpenStreetMap search + OSRM road routing

Figure 3.7: Live order-tracking timeline

3.8 Recipes and PDF Cards

The Recipes page lists 33 recipes in six categories – Burgers, Chaat & Street, Fries & Loaded, Snacks, Wraps & Sandwiches and Asian Fusion. Each recipe shows an embedded video, ingredients, cooking steps and a link to the related product, and the visitor can download a PDF recipe card.

3.9 Exports and Contact Modules

The Exports page draws a Leaflet world map with 46 country pins in 9 regions (North America, South America, Africa, Middle East, East Asia, South East Asia, Europe, Indian Subcontinent and Oceania). Tapping a pin shows a shipping route from the Mehsana plant, and the enquiry form lets a buyer ask for specifications, pricing and logistics. The Connect page lists the phone number, e-mail addresses and three offices, with a message form and quick answers.

Figure: Recipe and export enquiry workflows

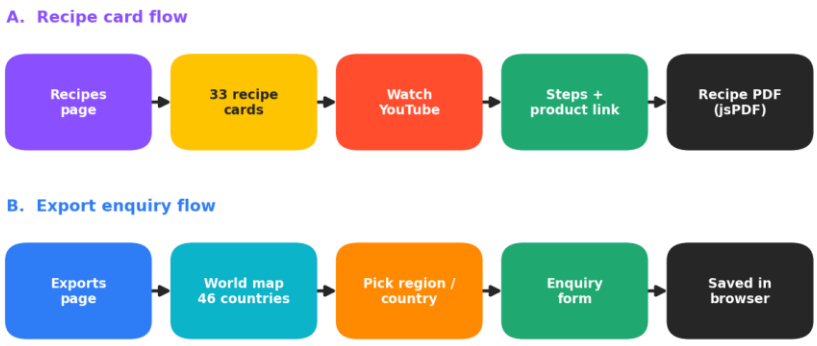


Figure 3.8: Recipe and export enquiry workflows

3.10 Seed to Shelf Process

This page explains how RT FOOD controls every step: certified seed potatoes, contract farms in North Gujarat's Banaskantha plus Punjab and Himachal Pradesh, farmer training, research trials, sub-zero cold storage, processing at the Mehsana plant, quality checks against FSSAI, BRC, Sedex and Kosher standards, and an unbroken cold chain.

Figure: Seed-to-shelf supply chain



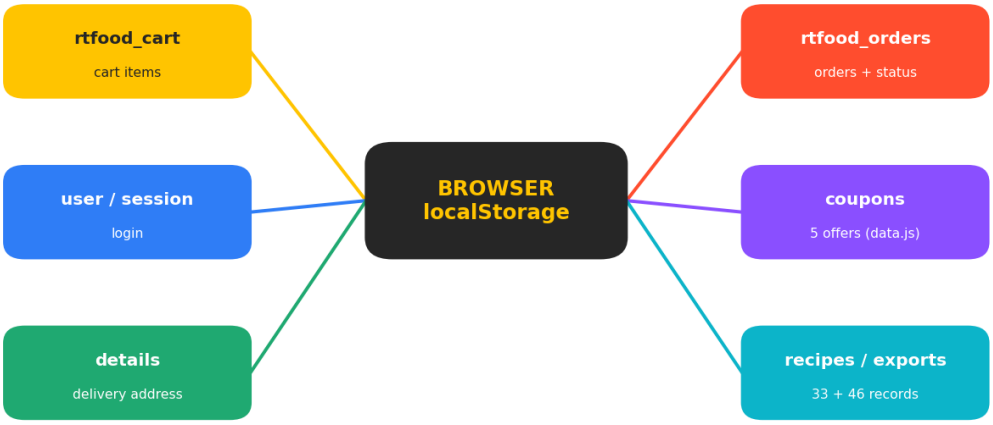
Stages follow the Seed to Shelf page of the website

Figure 3.9: Seed-to-shelf supply chain

3.11 Data Model

The website does not use a database. Product data, coupons, countries and recipes are fixed in `data.js`, while customer data is saved in localStorage. The diagram and table below show what is stored where.

Figure: Front-end data storage model



Open tabs stay in sync through the browser storage event

Figure 3.10: Front-end data storage model

Data	Where	Content
Products, coupons, GST, UPI ID	js/data.js (file)	Fixed business data edited by the owner
Cart	localStorage rtfood_cart	Product id, pack size, quantity
Login / session and details	localStorage	Name, phone, e-mail, delivery address
Orders	localStorage	Items, totals, payment method, status, time stamp
Recipes and export countries	js/data.js (file)	33 recipes, 46 countries with latitude and longitude

CHAPTER 4

SYSTEM IMPLEMENTATION

4.1 Project Structure

PROJECT FOLDER

```
RT-Food-Frozen-Snacks/
|-- index.html          Home
|-- about.html          About Us
|-- seedtoshelf.html    Seed to Shelf
|-- products.html       Products
|-- Exports.html        Exports
|-- Recipes.html        Recipes
|-- culture.html        Culture + CV upload
|-- connect.html        Contact
|-- order.html          Order Now
|-- css/style.css       one design system for every page
|-- assets/image/       logo, product and brand images
`-- js/                 data, main, shop, home, products, order,
                        pdf, logo, recipes, exports, forms
```

4.2 Application Shell

`main.js` is loaded on every page. It shows the “Heating up the fryer…” loader on the Home and Order pages, builds the shared header with the cart button, adds the footer, runs the cart drawer, shows toast messages and starts the scroll-reveal animations and number counters. Because of this shell each page file only needs to hold its own content.

4.3 Important Functions

Function	File	What it does
RT.price(product, size)	shop.js	Returns the price for a pack size using the size multiplier on the base price.
RT.productCard(p, size)	shop.js	Builds the HTML of one product card with chips, price and buttons.
RT.bindShop(root, onAdd)	shop.js	Handles size chips, Add and Buy Now clicks with one event listener.
RT.filterProducts(opts)	shop.js	Filters by text, category, tag and size and sorts the result.
RT.cart.add(id, size, qty)	main.js	Adds an item to the cart and saves it in localStorage.
applyCoupon(code)	order.js	Checks the code and minimum order and calculates the discount.
drawQR()	order.js	Creates the UPI payment QR with a five-minute refresh timer.
placeOrder()	order.js	Saves the order, clears the cart and opens the bill.
statusOf(order)	order.js	Converts elapsed time into the current delivery stage.
RT.pdf.bill(order) / recipe(r)	pdf.js	Generates the invoice and recipe-card PDF files.

4.4 Code Samples

Pack-size pricing (js/data.js)

```
RT.SIZES = { '200g': 0.42, '500g': 1, '1kg': 1.9, '2.5kg': 4.5,
             '1 Pack': 1, '2 Packs': 1.9, '4 Packs': 3.6 };

RT.price = (prod, size) => Math.round(prod.base * RT.SIZES[size]);
```

Coupon definitions (js/data.js)

```
RT.COUPONS = {
  WELCOMERT: { type: 'pct',  value: 10, max: 150, min: 0   },
  FLAT100:   { type: 'flat', value: 100,          min: 699 },
  FRYDAY:    { type: 'pct',  value: 15, max: 200, min: 999 },
  CRISPY50:  { type: 'flat', value: 50,           min: 349 },
  FREESHIP:  { type: 'ship', value: 0,            min: 0   }
};
```

Compressed tracking timeline (js/order.js)

```
const TRACK = { confirm: 10, pack: 28, pick: 42, ride: 100 };  // seconds
```

4.5 PDF and QR Implementation

jsPDF draws the invoice and the recipe cards page by page in the browser: logo, order number, customer details, item table, GST and total. Because the logo must be drawn inside the PDF, it is stored as a base64 string in `logo.js`. The UPI payment QR code is created by qrcode.js from the UPI ID, the order amount and a note, so a customer can scan it with any UPI app.

4.6 Running and Deployment

1. Download or clone the repository from GitHub.
2. Open a terminal in the folder and run `python -m http.server 8000`.
3. Visit `http://localhost:8000` in a browser (internet is needed for fonts, maps, jsPDF, QR code and videos).
4. To publish, push to GitHub and enable GitHub Pages for the main branch – the site goes live as a static website.

4.7 Security and Validation Notes

- This is an educational front-end demo: accounts are stored only in the browser and are **not secure**.
- Payments are simulated; the UPI ID, GST number and rates in `data.js` are placeholders to be replaced before any real use.
- The details form checks a 10-digit mobile number (starting 6–9), a 6-digit PIN code and a full address; card payments check the card number with the Luhn rule, expiry, CVV and name. Real validation must still be added on a server.
- The footer states that the website is for educational purposes only.

CHAPTER 5

SOFTWARE TESTING

5.1 Test Approach

The website was tested with automated browser checks. A headless Chromium browser (Playwright) opened a local copy of the project files – the same files that are published on GitHub Pages – and ran 16 checks. External services such as Google Fonts, map tiles, YouTube and CDN libraries were blocked during the run, so the checks cover the project's own HTML, CSS and JavaScript.

> **Result**
> 16 of 16 checks passed.

5.2 Page Loading Tests

Each page was opened and checked for JavaScript errors and for the presence of the shared header.

#	Test case	Expected result	Result
1	index.html loads without JavaScript errors	Page opens, header present, no script errors	PASS
2	about.html loads without JavaScript errors	Page opens, header present, no script errors	PASS
3	products.html loads without JavaScript errors	Page opens, header present, no script errors	PASS
4	order.html loads without JavaScript errors	Page opens, header present, no script errors	PASS
5	Recipes.html loads without JavaScript errors	Page opens, header present, no script errors	PASS
6	Exports.html loads without JavaScript errors	Page opens, header present, no script errors	PASS
7	connect.html loads without JavaScript errors	Page opens, header present, no script errors	PASS
8	culture.html loads without JavaScript errors	Page opens, header present, no script errors	PASS
9	seedtoshelf.html loads without JavaScript errors	Page opens, header present, no script errors	PASS

5.3 Catalogue, Filters and Cart Tests

#	Test case	Expected result	Result
10	Products page renders all product cards	52 product cards displayed	PASS
11	Add to cart stores item in localStorage	Key rtfood_cart is created	PASS
12	Search for "hash" returns only matching products	7 of 52 products shown	PASS
13	Category filter "French Fries" returns that category	11 products shown	PASS

#	Test case	Expected result	Result
14	Price changes with pack size (1 kg higher than 500 g)	1 kg price greater than 500 g price	PASS

5.4 Order Page and Responsive Tests

#	Test case	Expected result	Result
15	Order page shows the 7-step progress tracker	Login to Track steps visible	PASS
16	Mobile (390 px): no visible horizontal scrolling	Body overflow hidden, menu off-canvas	PASS

Note on the mobile check: at 390 px width the page's scroll width is 416 px because the slide-in navigation menu is positioned off-screen. The body uses `overflow-x: hidden`, so no horizontal scrolling is visible to the user, which is the intended behaviour.

5.5 Test Summary and Limits

16	16	0
CHECKS RUN	PASSED	FAILED

Not covered by these automated checks: scanning the UPI QR with a real app, live map tiles and OSRM routing, YouTube playback, file upload for the CV form and real payment processing. These need an internet connection or external services and were not part of this test run.

CHAPTER 6

SCREENSHOTS AND WALKTHROUGH

The screenshots below walk through the website and the complete order flow. They were captured from the project source code and are framed with the live address of the website: https://shahrishabh1513-jsk.github.io/RT-Food-Frozen-Snacks/

Maps and embedded videos load from external services on the live site, so they may look empty in a few captures.

6.1 Home Page

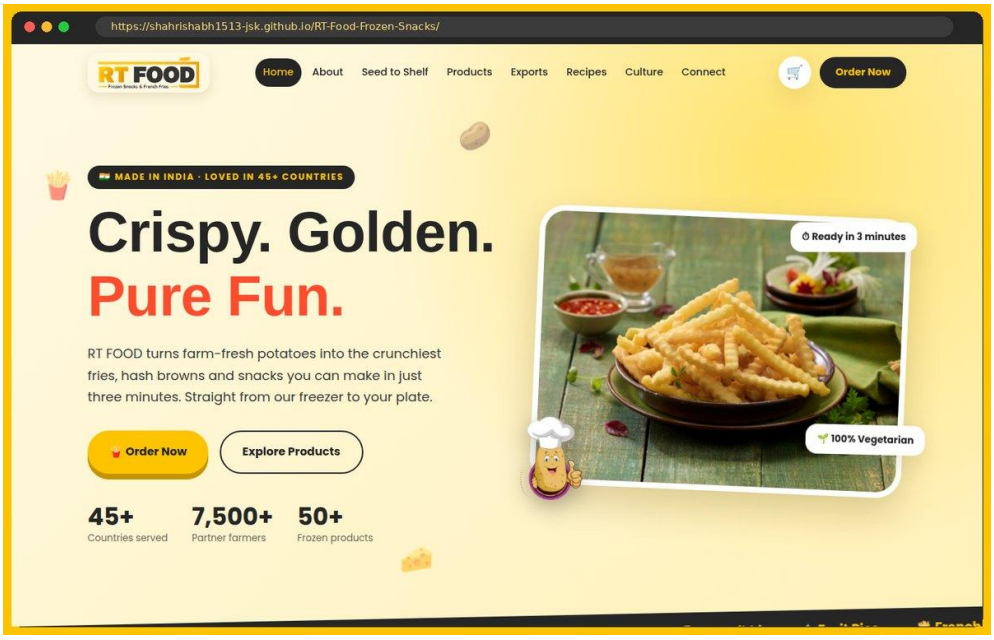


Figure 6.1: Home page – animated hero with brand call-to-action

The hero shows the headline “Crispy. Golden. Pure Fun.”, the Order Now and Explore Products buttons, and counters for countries served, partner farmers and frozen products.

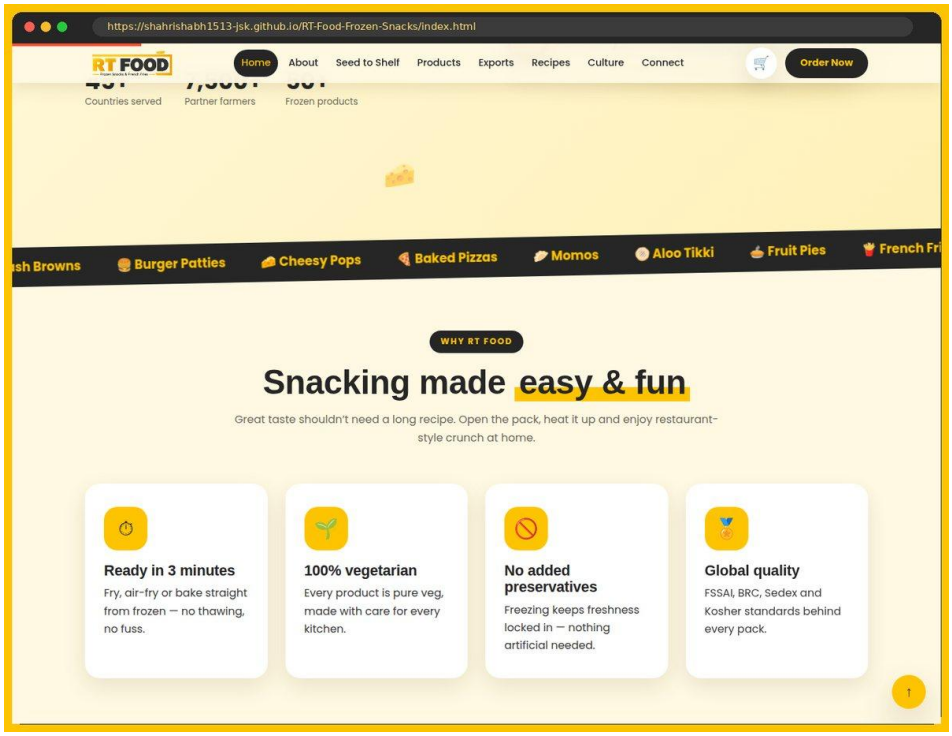


Figure 6.2: Home page – bestsellers and category strip

6.2 About Page

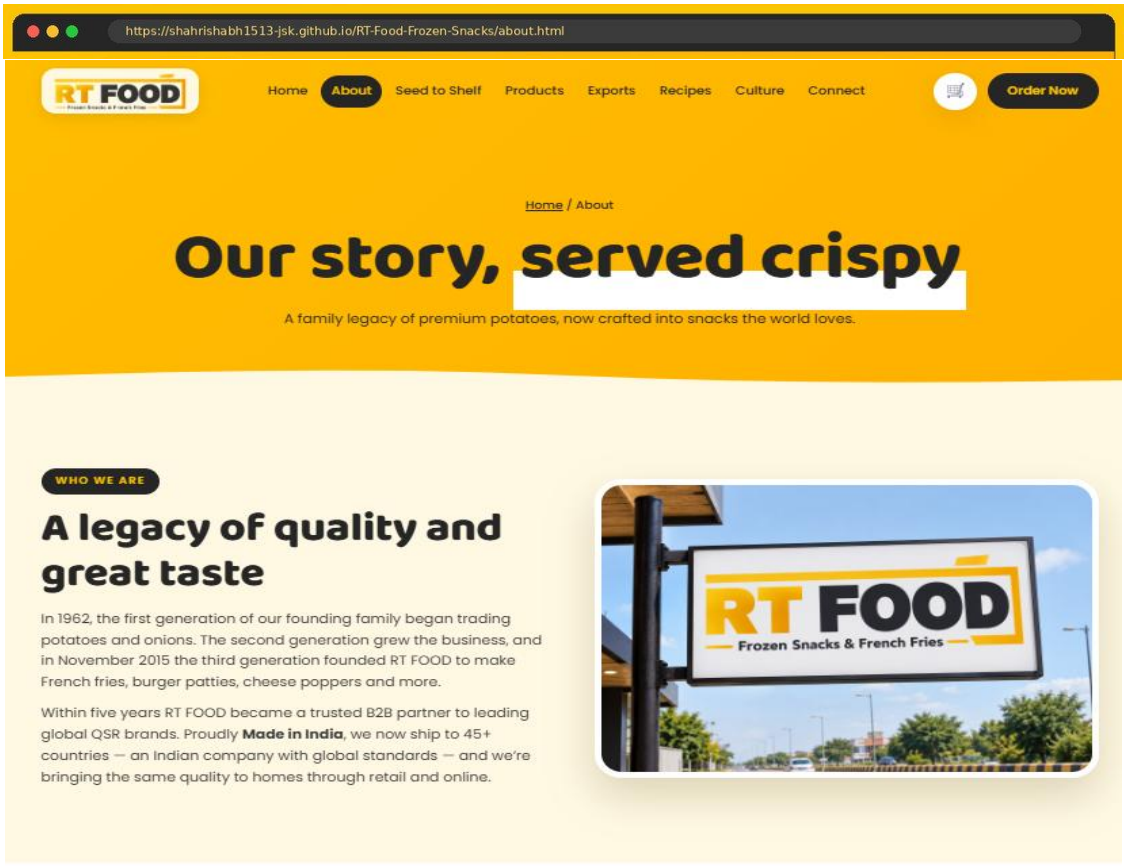


Figure 6.3: About – company story

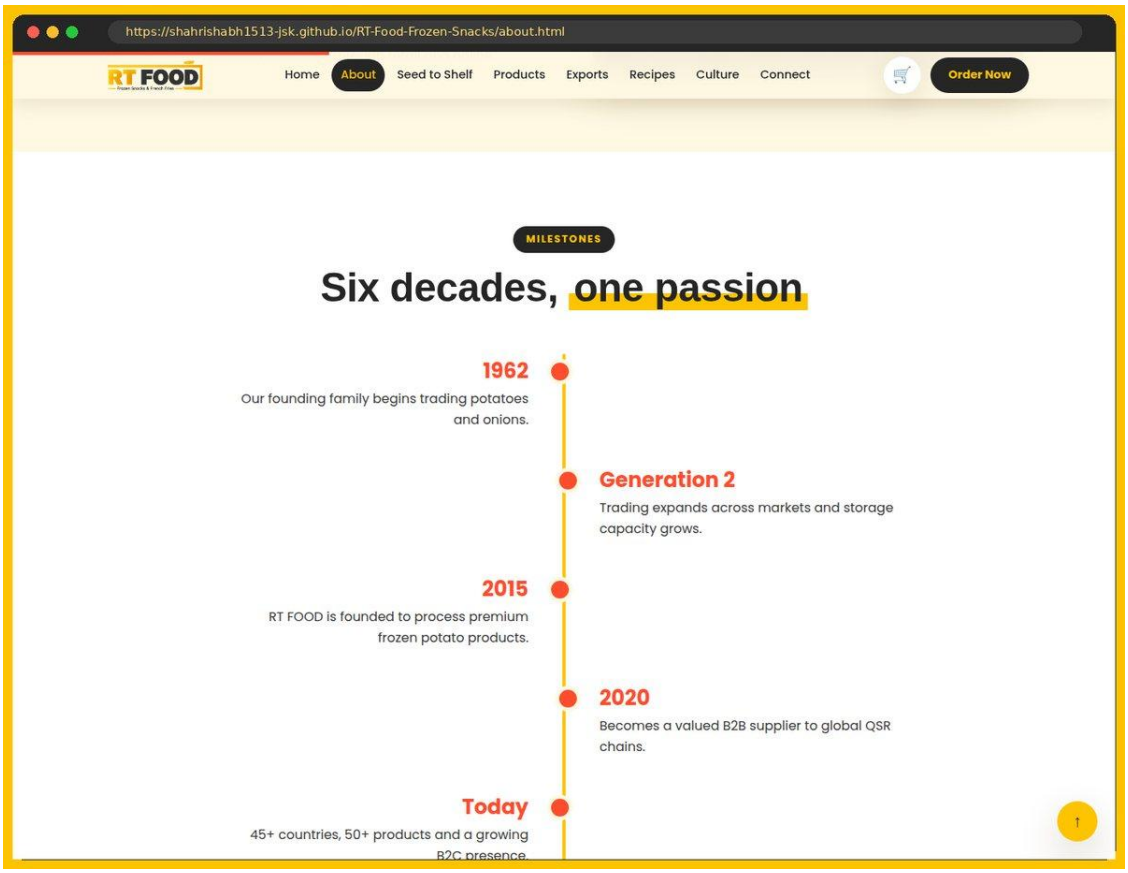


Figure 6.4: About – milestones, mission and vision

6.3 Seed to Shelf Page

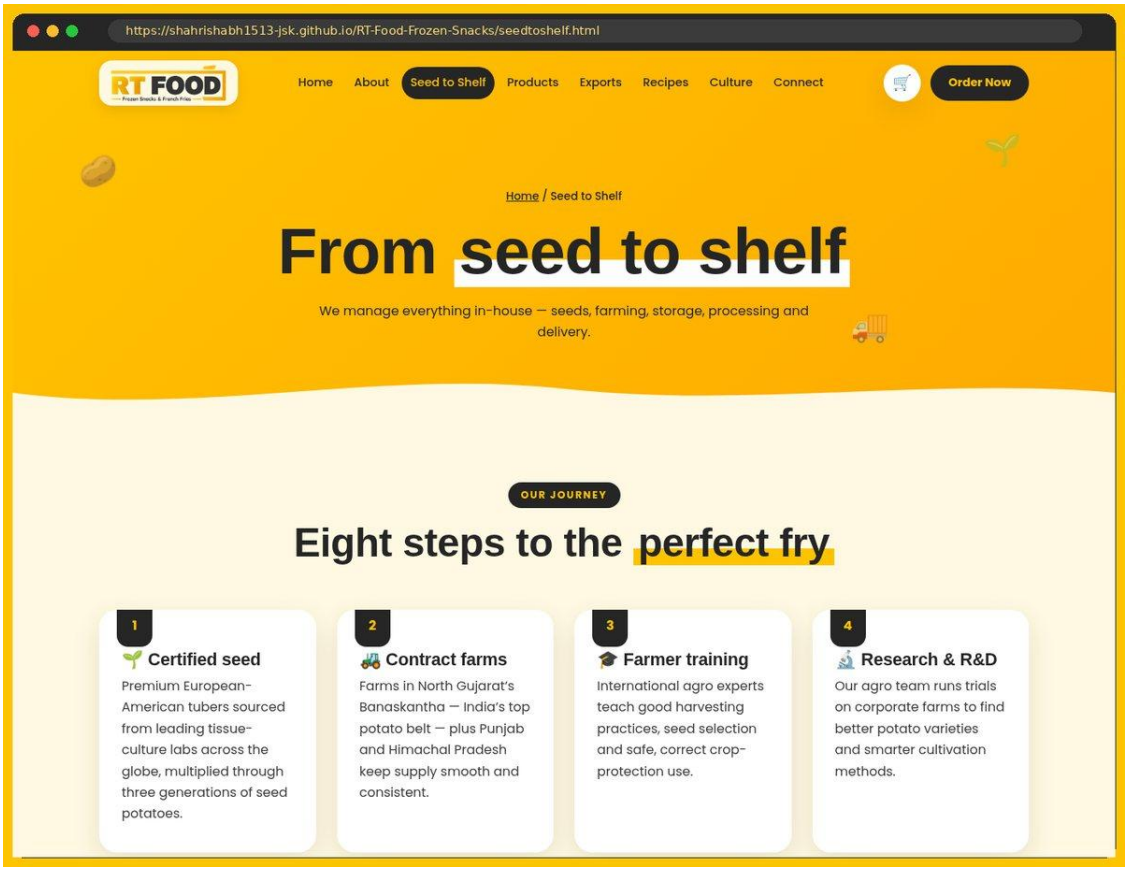


Figure 6.5: Seed to Shelf – page header

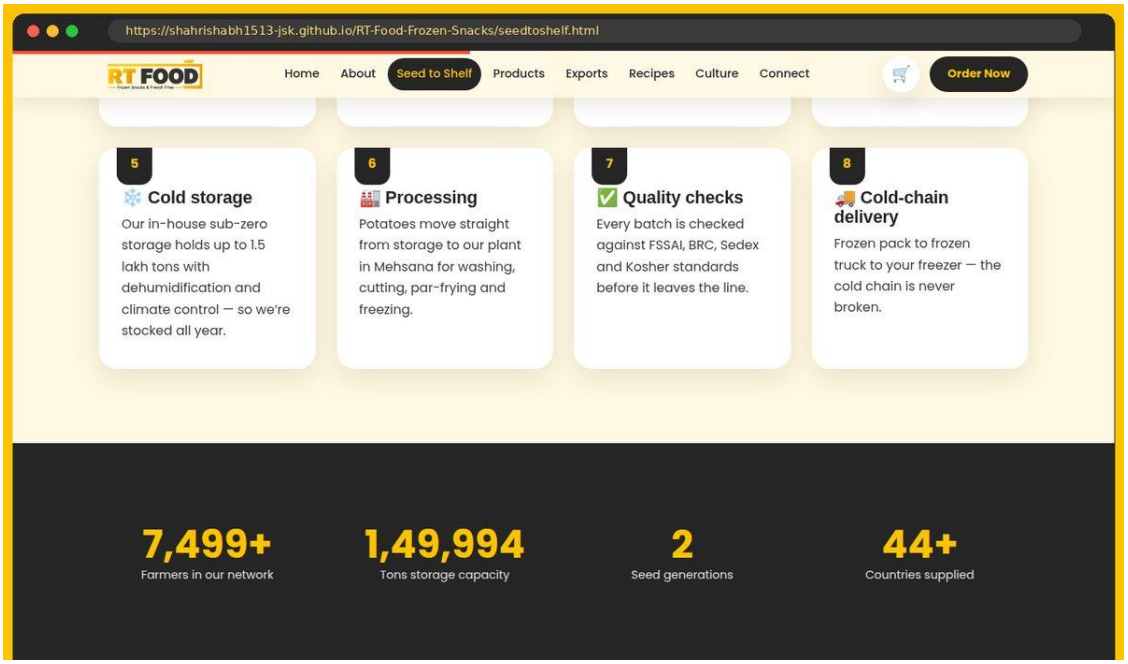


Figure 6.6: Seed to Shelf – eight steps of the supply chain

6.4 Products Page

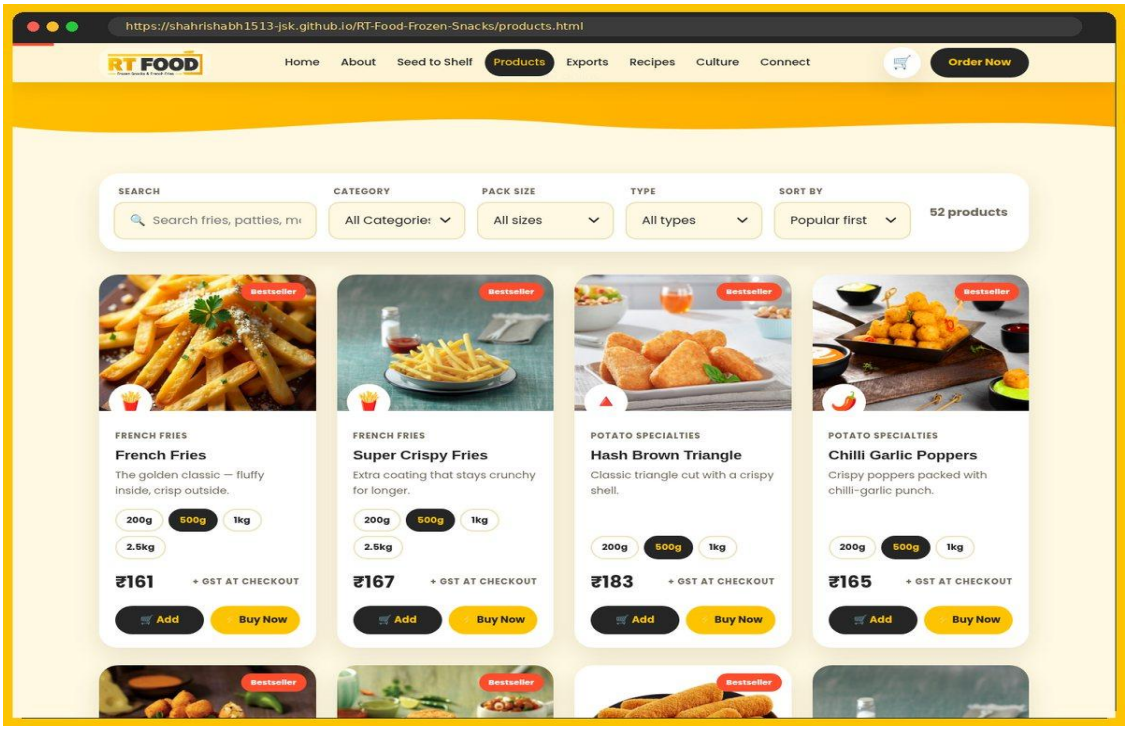


Figure 6.7: Products – filters and product cards with pack-size chips

6.5 Order Flow Walkthrough

The following seven screens show a complete demonstration order, from login to live tracking.

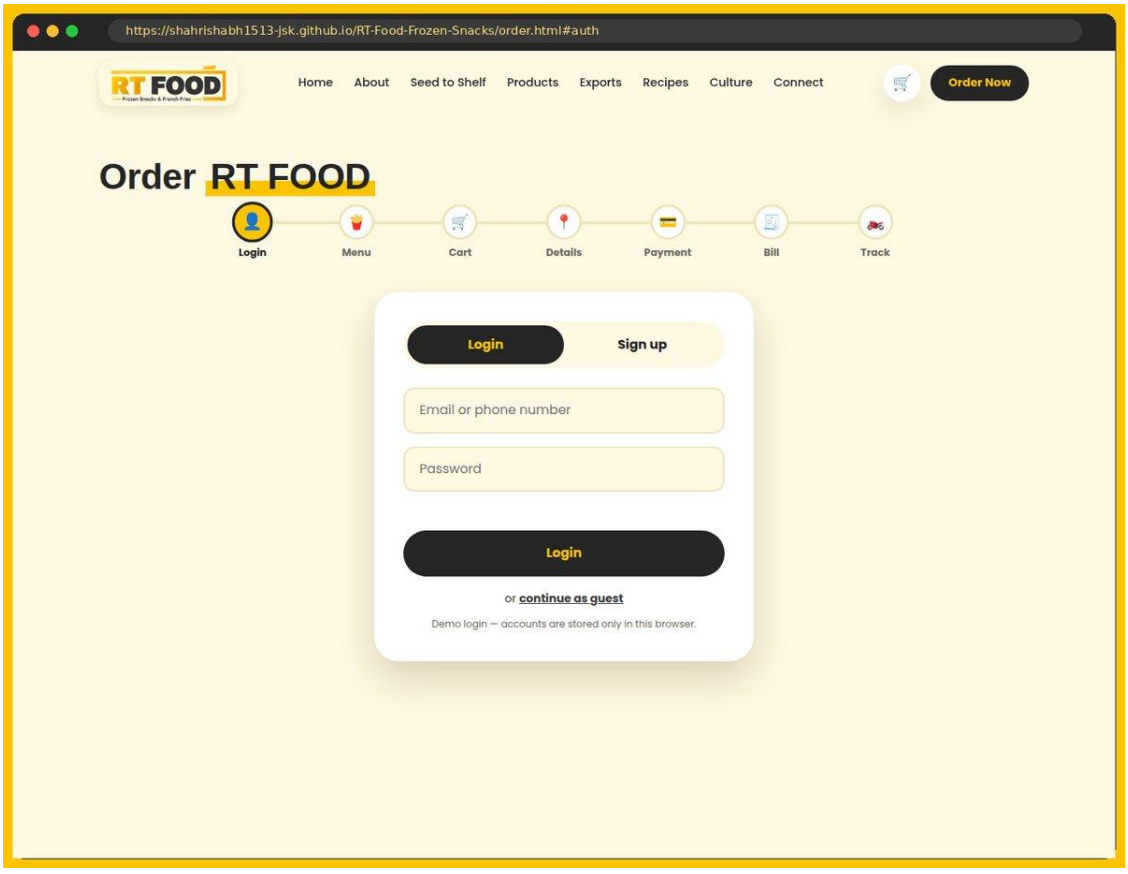


Figure 6.8: Step 1 – Login, sign-up or guest

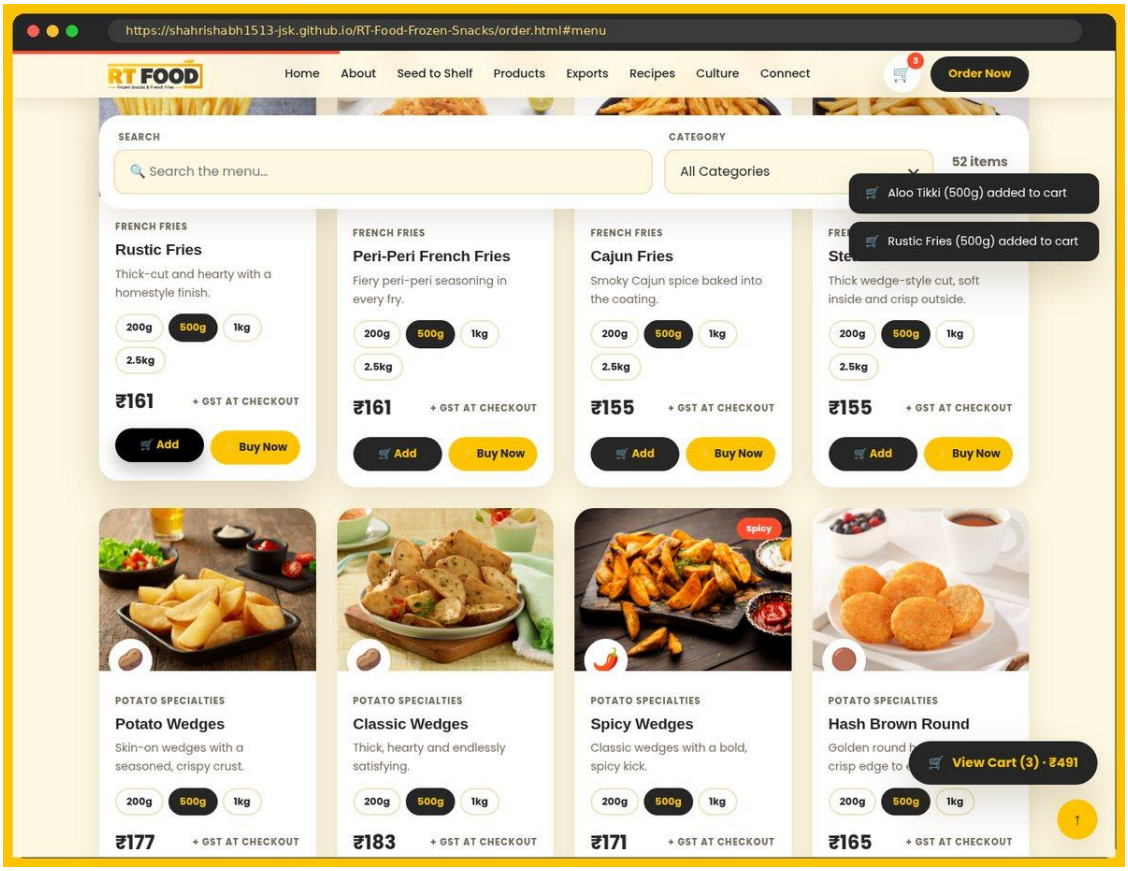


Figure 6.9: Step 2 – Menu with Add buttons and a View Cart bar

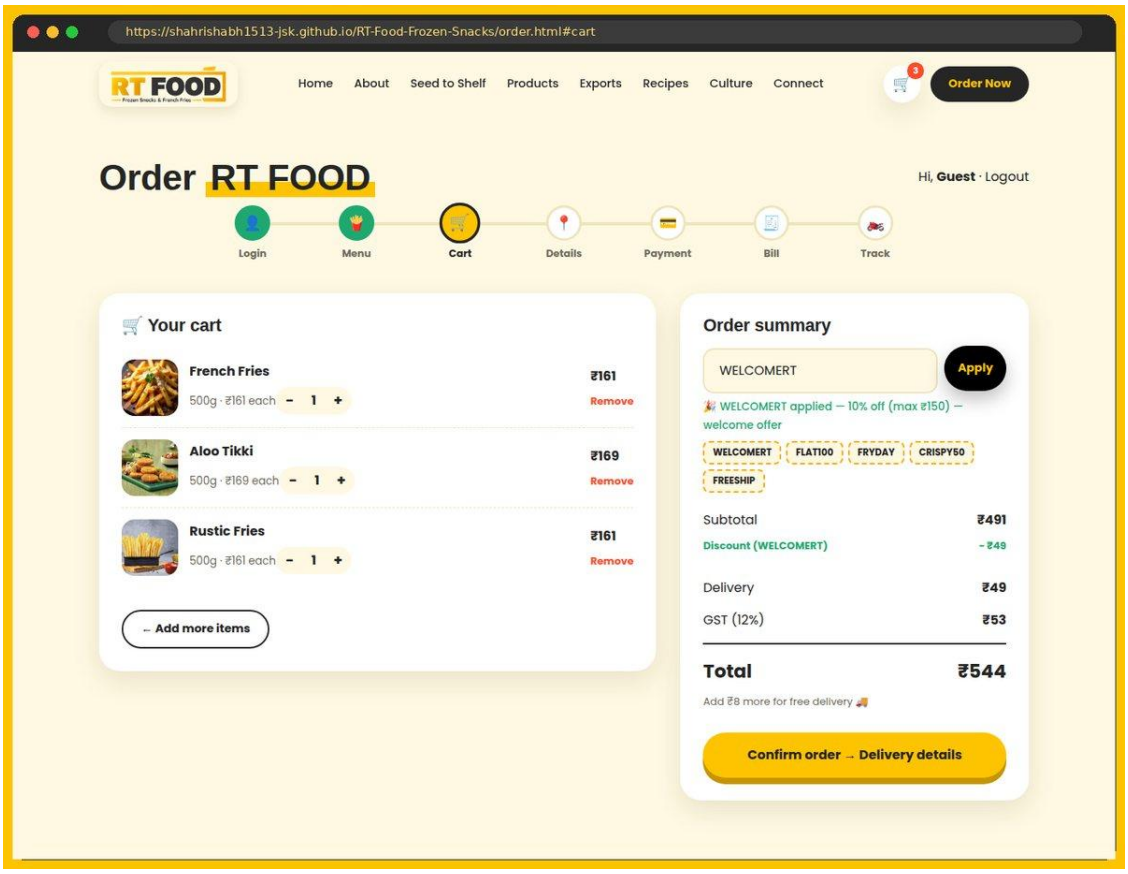


Figure 6.10: Step 3 – Cart with the WELCOMERT coupon applied

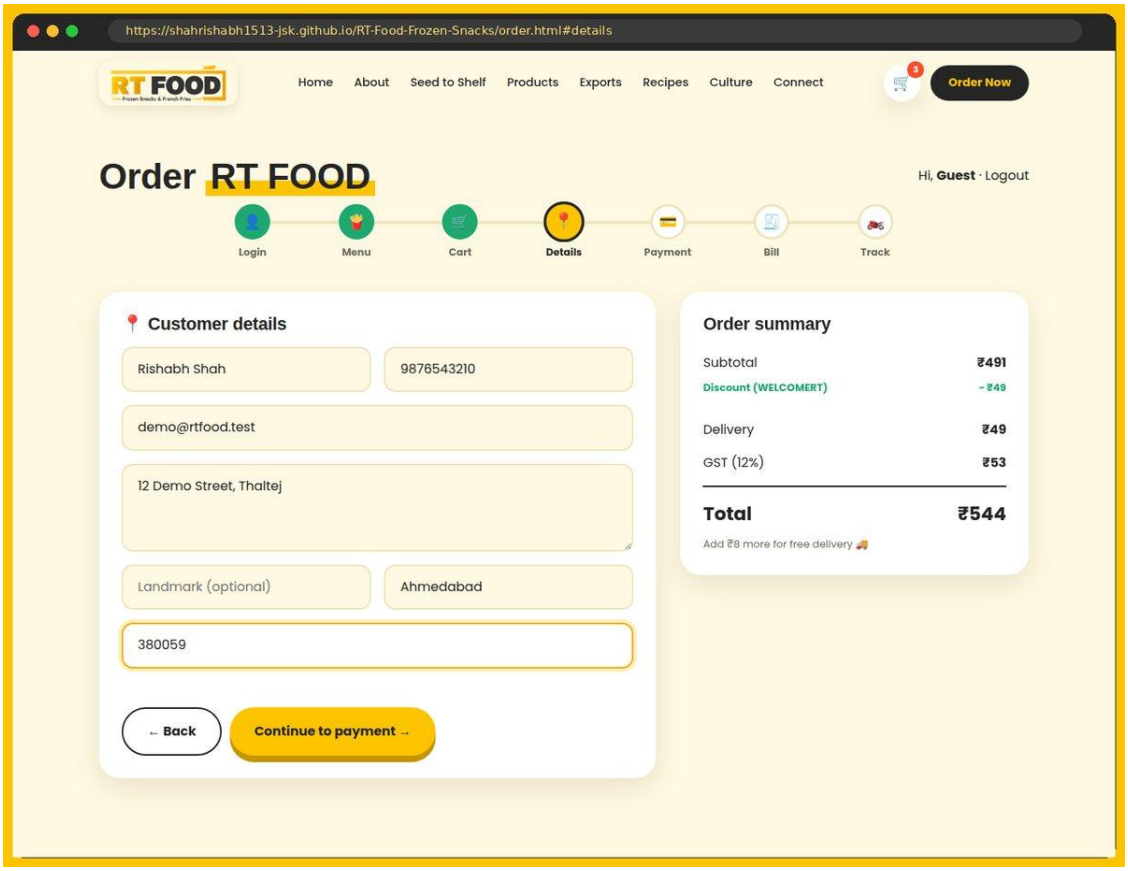


Figure 6.11: Step 4 – Customer details with validation

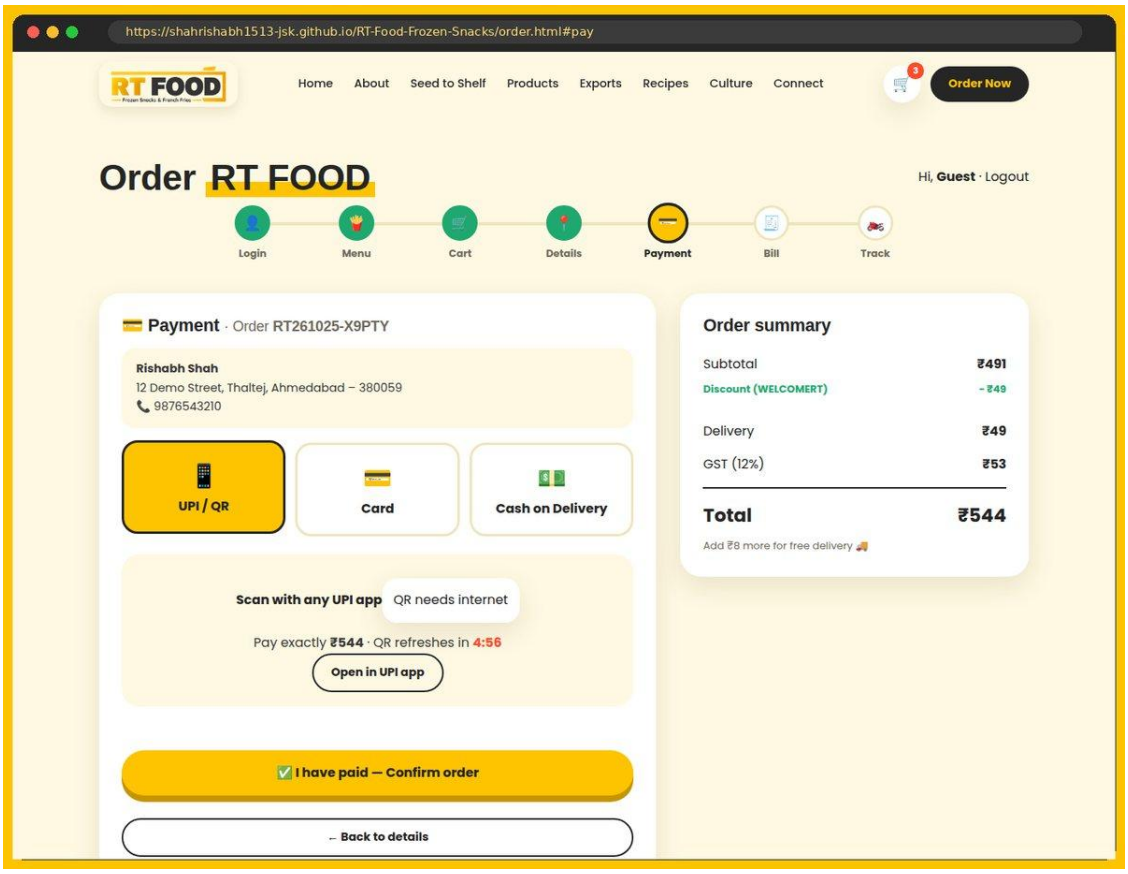


Figure 6.12: Step 5 – Payment (UPI / Card / Cash on Delivery)

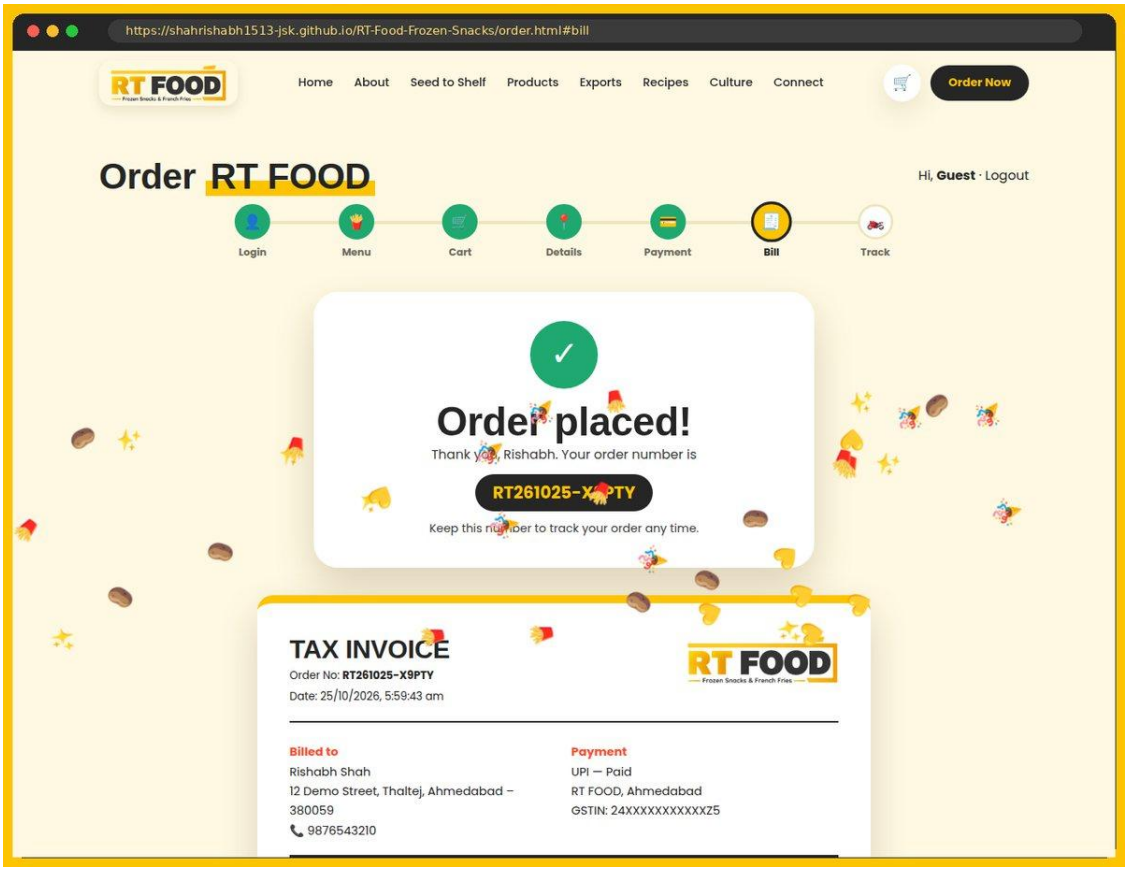


Figure 6.13: Step 6 – Order placed and tax invoice

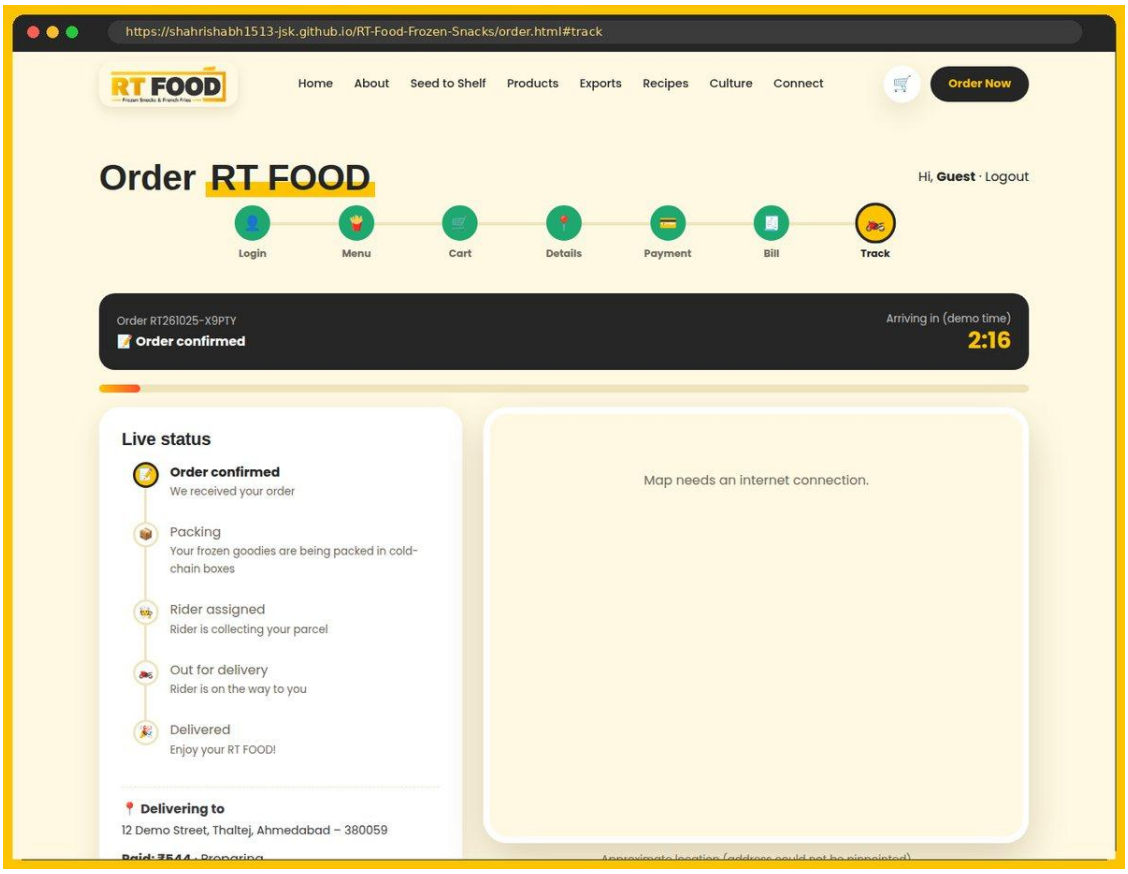


Figure 6.14: Step 7 – Live status (the map needs internet to load tiles)

6.6 Exports Page

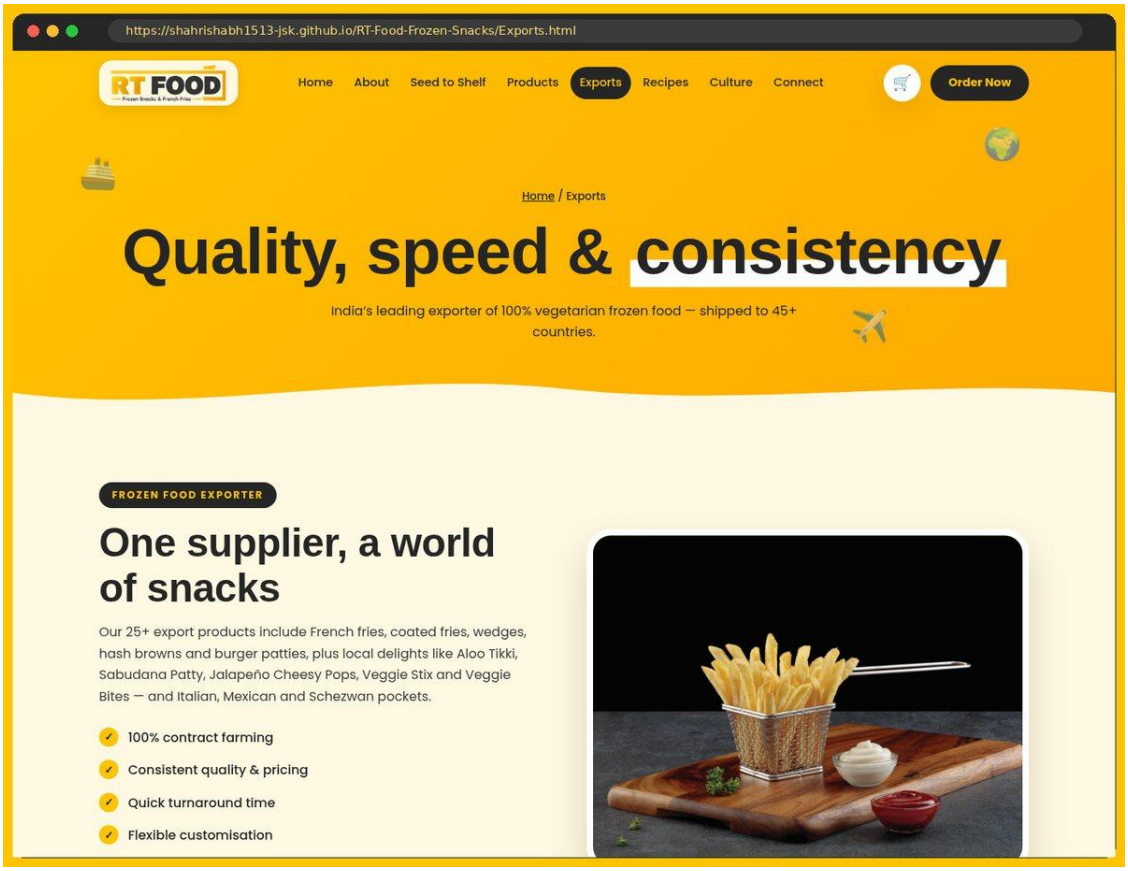


Figure 6.15: Exports – page header

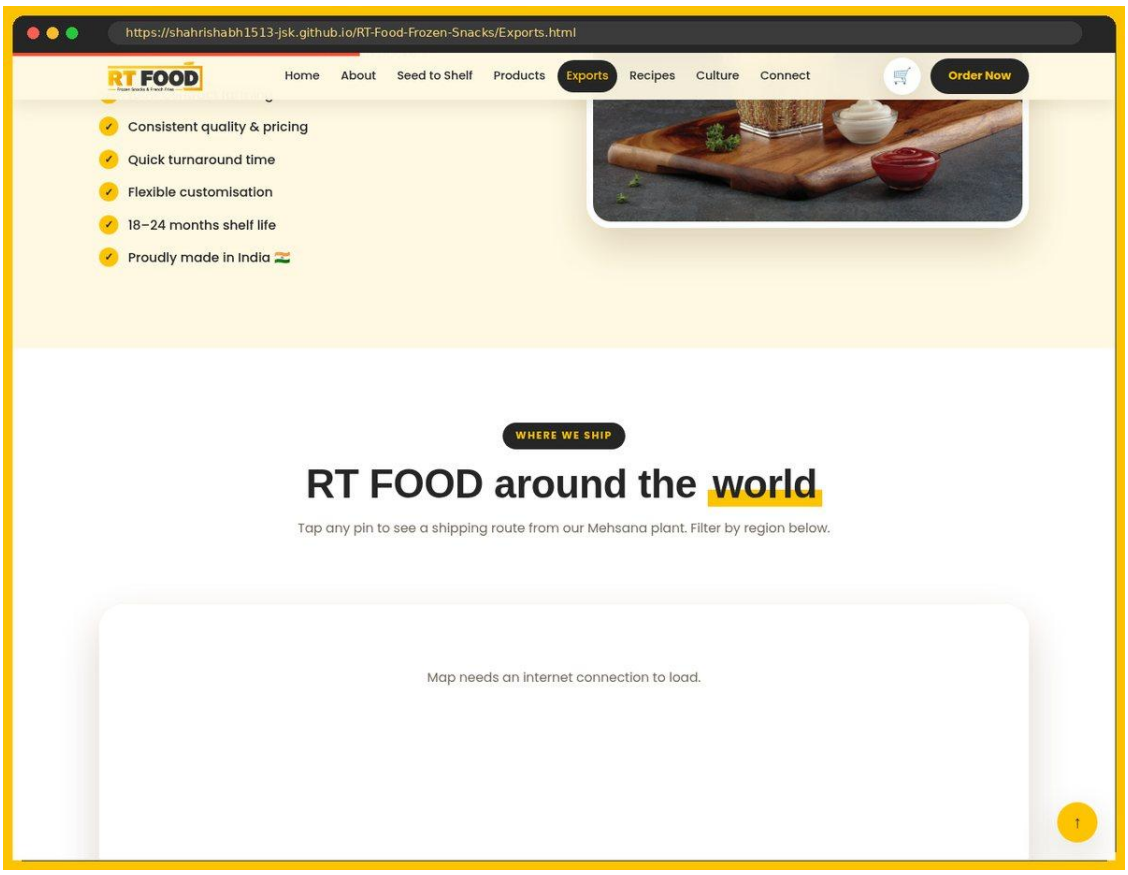


Figure 6.16: Exports – map section and enquiry form

6.7 Recipes Page

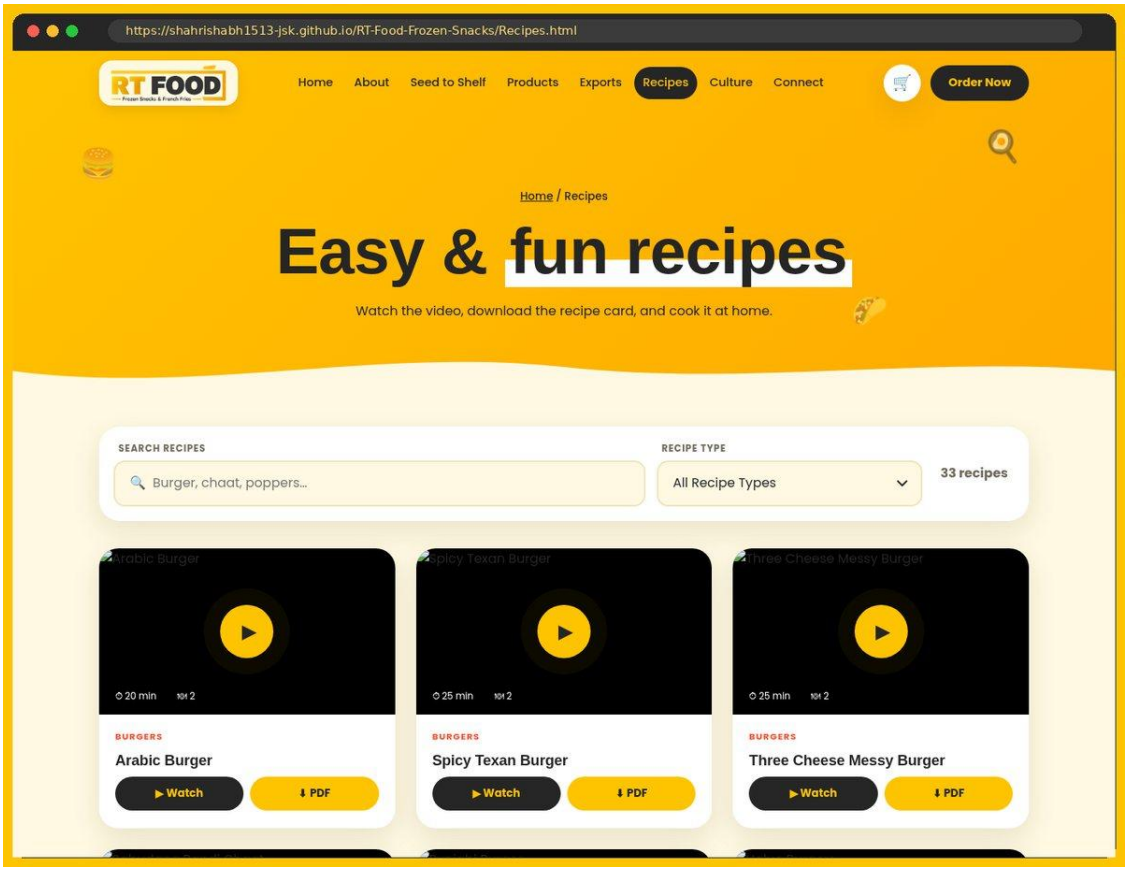


Figure 6.17: Recipes – 33 video recipes with PDF cards

6.8 Culture Page

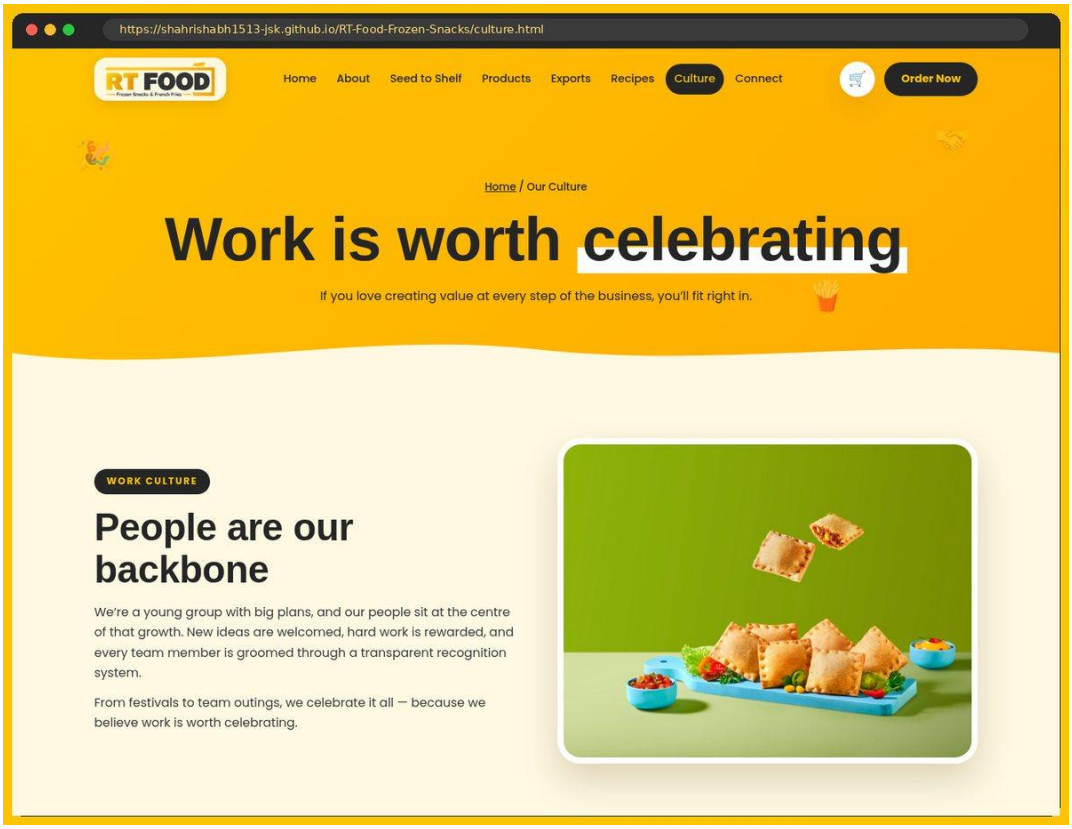


Figure 6.18: Culture – values and careers form

6.9 Connect Page

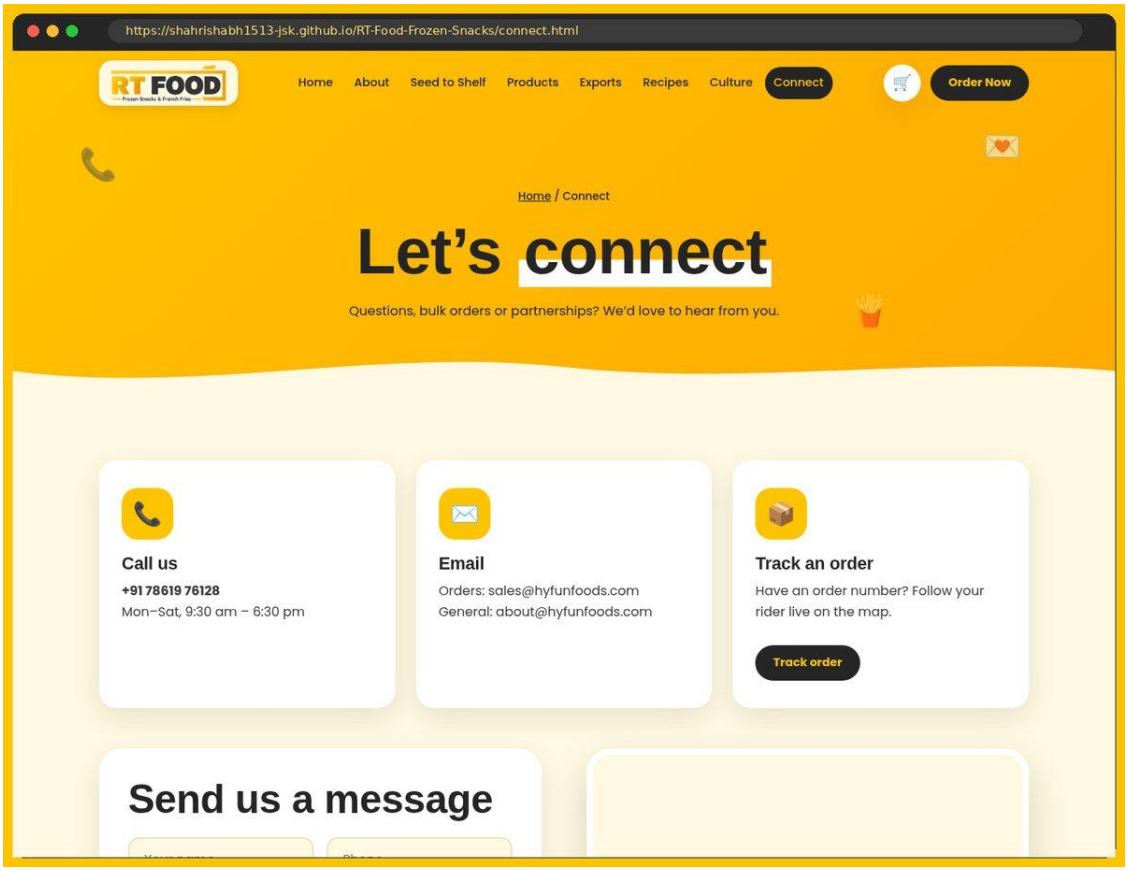


Figure 6.19: Connect – contact details

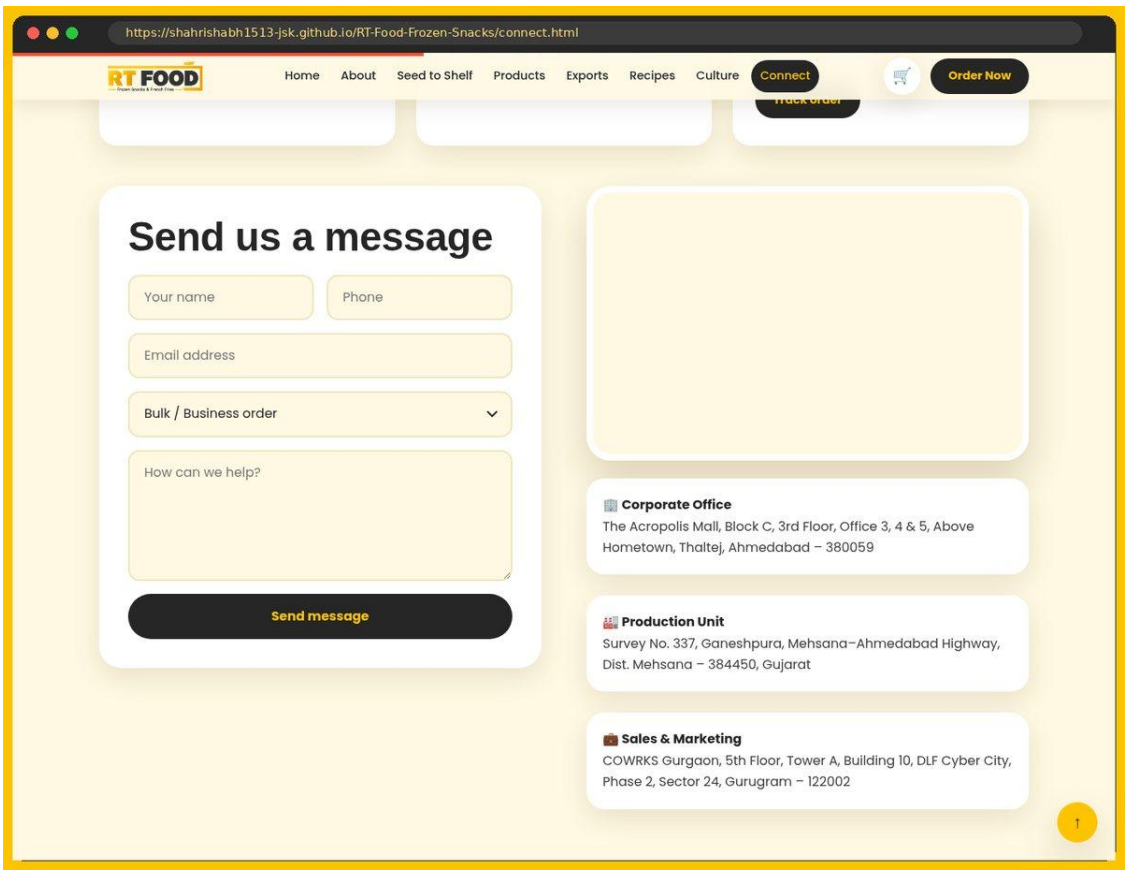


Figure 6.20: Connect – offices, message form and FAQ

6.10 Mobile View

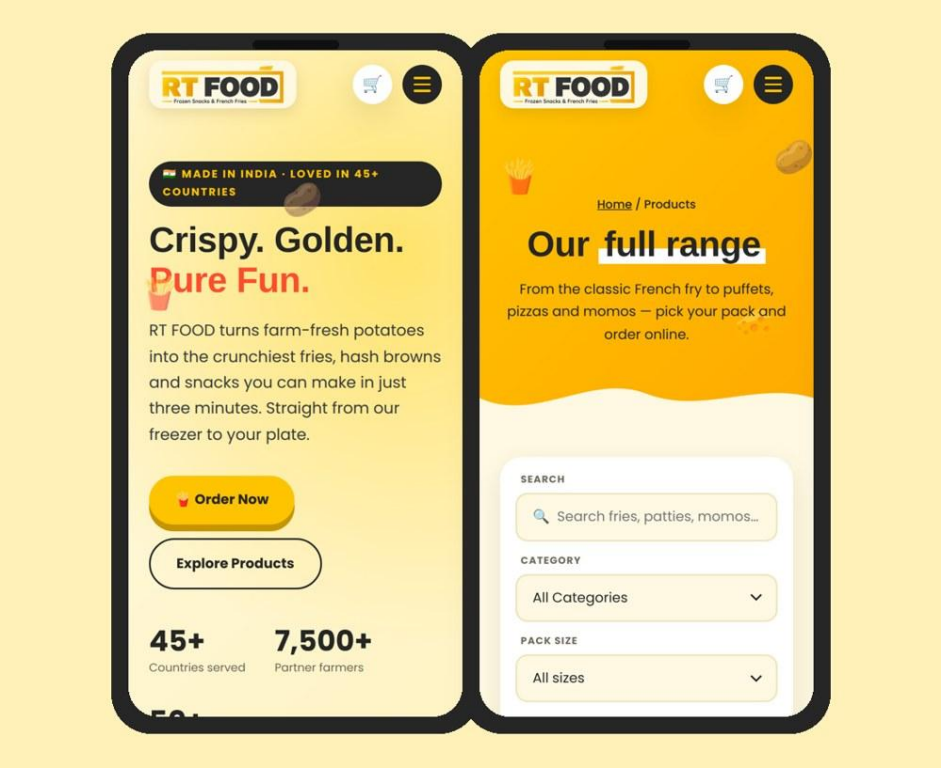


Figure 6.21: Home and Products pages on a 390 px phone screen – fully responsive

CHAPTER 7

LEARNING OUTCOMES

7.1 Skills and Knowledge Gained

- Designing a complete multi-page website with one reusable design system and shared components.
- Writing clean, modular Vanilla JavaScript with a single `RT` namespace and a data-driven catalogue.
- Building an e-commerce flow – cart, coupons, taxes, delivery fee, payment and billing.
- Using localStorage and the storage event for persistent, multi-tab state without a backend.
- Integrating Leaflet maps, OpenStreetMap geocoding and OSRM routing for live tracking.
- Generating PDF documents and QR codes directly in the browser.
- Creating responsive layouts, CSS animations and an attractive visual identity.
- Testing a web application with automated browser checks and publishing it on GitHub Pages.

7.2 Challenges and Solutions

Challenge	Solution
Managing 52 products and their prices in many places.	Moved everything into one data file (`data.js`) and generated all cards, filters and prices from it.
Different pack sizes for different product types.	Added a size multiplier table and a rule that gives weights to fries and packs to baked snacks.
Showing realistic delivery without a backend.	Used OSRM road routes and a compressed timer so a rider moves along a real path in the demo.
Putting the logo inside generated PDFs.	Stored the logo as base64 in `logo.js` so jsPDF can draw it.
Keeping nine pages consistent.	Created one stylesheet and one shared shell (`main.js`) for header, footer, cart and loader.

7.3 Limitations

- No real backend or database – data exists only in the browser of each user.
- Payments are simulated and accounts are not secure; passwords are not hashed.
- Maps, fonts, libraries and videos need an internet connection.
- Rider movement and order status are simulated, not real-time dispatch data.
- Recipe steps are written guides based on the video titles and should be verified in a test kitchen.

CHAPTER 8

CONCLUSION

8.1 Conclusion

RT FOOD successfully shows how a complete, attractive and useful brand website can be built with only HTML, CSS and JavaScript. It combines storytelling pages, a searchable 52-product catalogue, a seven-step ordering journey with coupons, GST, payment options, PDF billing and live rider tracking, together with export, recipe, careers and contact features.

The project achieved its objectives: a colourful and responsive design, a data-driven code structure, a working demonstration of the full e-commerce workflow and a live deployment on GitHub Pages, backed by 16 passing automated checks. It also gave practical experience in front-end architecture, state management, maps, document generation and testing.

8.2 Future Enhancements

Enhancement	Benefit
Backend with a database (Node.js / Firebase)	Real accounts, shared orders and an admin dashboard.
Real payment gateway (Razorpay / UPI intent)	Accept real payments securely.
Password hashing and authentication tokens	Secure login for customers.
E-mail and WhatsApp order notifications	Order confirmation and delivery updates.
Real rider app and live GPS	True live tracking instead of simulation.
Wishlist, reviews and loyalty points	Higher engagement and repeat orders.
Multi-language support (Hindi, Gujarati)	Reach more customers.
Progressive Web App (offline and installable)	App-like experience on mobile.

CHAPTER 9

REFERENCES AND GLOSSARY

9.1 References

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2. MDN Web Docs – Web Storage API and the storage event. https://developer.mozilla.org/docs/Web/API/Web_Storage_API
3. Leaflet – an open-source JavaScript library for interactive maps. https://leafletjs.com
4. jsPDF – client-side PDF generation. https://github.com/parallax/jsPDF
5. OpenStreetMap – open map data and tiles. https://www.openstreetmap.org
6. OSRM – Open Source Routing Machine. https://project-osrm.org
7. GitHub Pages documentation. https://docs.github.com/pages
8. Project source code and live website. https://github.com/shahrishabh1513-jsk/RT-Food-Frozen-Snacks

9.2 Glossary

Term	Meaning
Frozen snack	Ready-to-cook food kept frozen, such as fries, hash browns, patties and pizzas.
QSR	Quick Service Restaurant – a fast-food chain that buys frozen products in bulk.
GST	Goods and Services Tax added to the order total (set to 12% in this project).
Coupon	A code that gives a percentage, flat or free-delivery discount.
localStorage	Browser storage that keeps data on the user's device after the page is closed.
Geocoding	Converting an address into map coordinates.
OSRM	A routing engine that finds the road route between two map points.
jsPDF	A JavaScript library that creates PDF files in the browser.
UPI	Unified Payments Interface – India's instant payment system used by QR codes.
Cold chain	An unbroken chain of freezing and refrigerated storage and transport.
GitHub Pages	Free hosting of static websites directly from a GitHub repository.

LET'S CONNECT

Thank you for reading this report. Open the live project, explore the code or get in touch.

LIVE PROJECT →
Open the RT FOOD website

PROJECT SOURCE CODE →
GitHub repository of this project

GITHUB PROFILE →
github.com/shahrishabh1513-jsk

LINKEDIN →
Rishabh Alpeshabhai Shah

SCAN TO OPEN
the live website

LINKS AT A GLANCE

Live Project: https://shahrishabh1513-jsk.github.io/RT-Food-Frozen-Snacks/
Project Link: https://github.com/shahrishabh1513-jsk/RT-Food-Frozen-Snacks
GitHub: https://github.com/shahrishabh1513-jsk
LinkedIn: https://www.linkedin.com/in/rishabh-alpeshabhai-shah-91b9072a6/

DESIGNED AND DEVELOPED BY

SHAH RISHABH ALPESHBHAI

Developer ID RT1504 · RT FOOD – Frozen Snacks & French Fries